```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[NG_FFCutsFits,CutFitFileNG] = recover_selected_cuts('NG_FFCutsFits', ...
    'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts');
```

```
fn = fieldnames(NG_FFCutsFits);
nf = 11
```

```
nf =
11
```

```
%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei400_off110NGminFF'
```

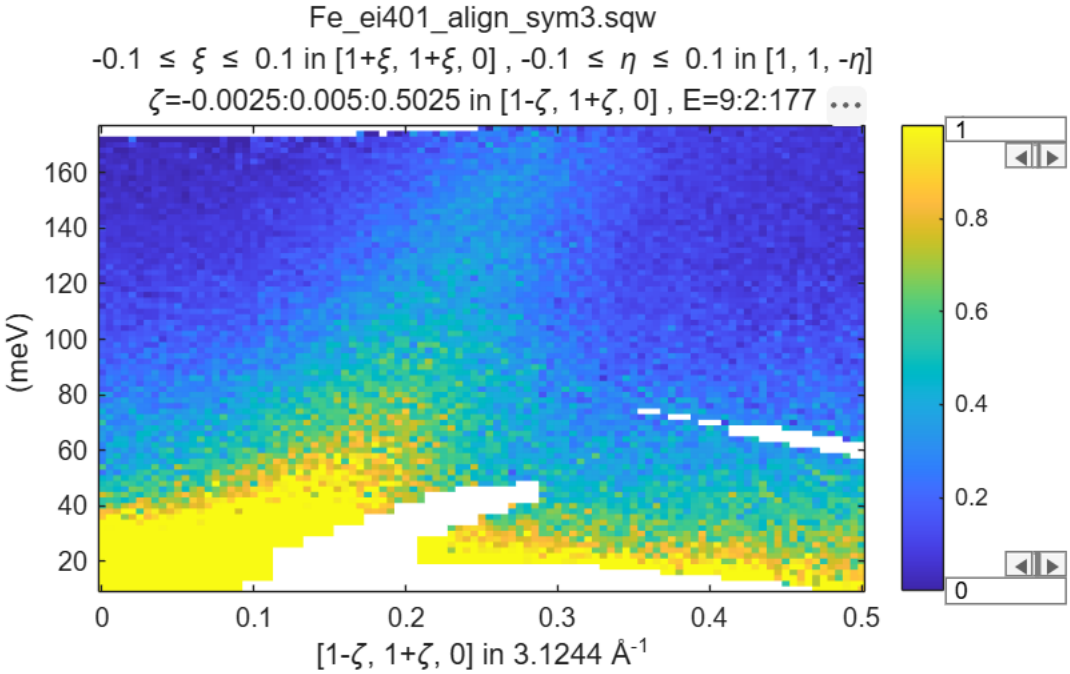
```
selectCuts =
'Ei400_off110NGminFF'
```

```
if ismember(selectCuts(1:3),{'msk','dat'})||...
        contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei400_off110NGminFF
```

```
cuts_list = NG_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
    cuts_list{j} = cuts_list{j}.set_instrument(mi);
    cuts_list{j} = cuts_list{j}.set_sample(sample);
    plot(cuts_list{j}); lz 0 1; keep_figure;
end
```




```
dir_name  = "NG";

dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 380;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
NG_FFCutsFits.(selectCuts) = cuts_list;
```

```
NG_FFCutsFits = fit_multicutsFF_along_direction(...
    NG_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

```
*****************************
**** En = 45±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 1.8865s    Reduced Chi^2 = 25.443
 Free parameter values:
           10         0.7917             34         0.6452
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 5.0592s    Reduced Chi^2 = 2.3212      Levenberg-Marquardt = 0.1
 Free parameter values:
        16.82         0.7552          21.76         0.4812
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 8.1127s    Reduced Chi^2 = 3.0007      Levenberg-Marquardt = 0.01
```

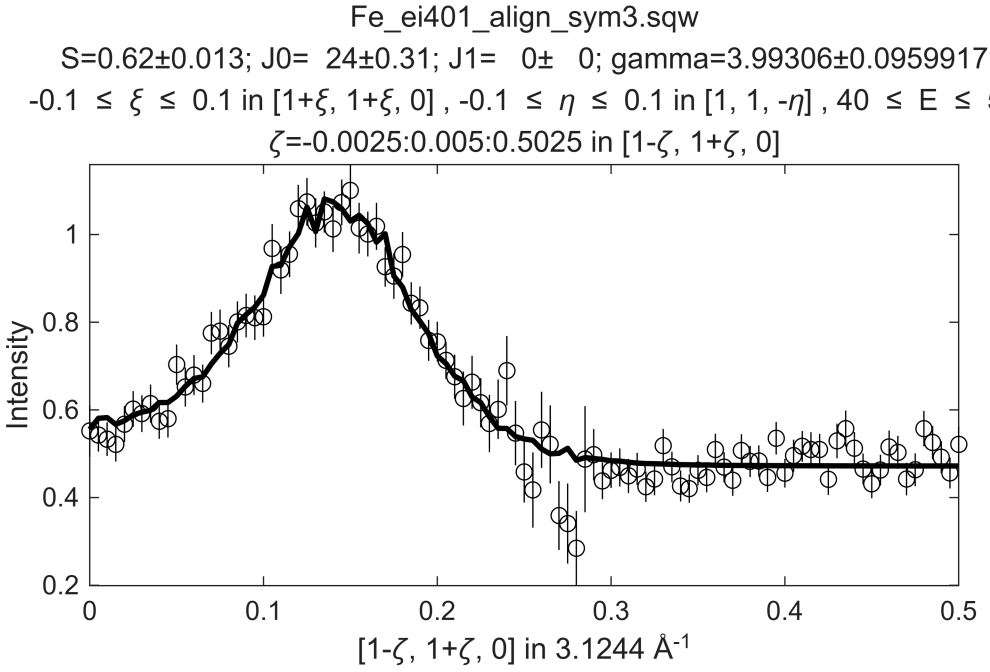
```
 Free parameter values:
       -4.652         0.4155          22.47         0.4845
 Total time = 8.9062s    Reduced Chi^2 = 0.95894      Levenberg-Marquardt = 0.1
 Free parameter values:
        2.711         0.5759           23.2         0.4734
------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 11.9863s    Reduced Chi^2 = 0.91064      Levenberg-Marquardt = 0.01
 Free parameter values:
        3.956         0.6115          23.51         0.4715
------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 14.9596s    Reduced Chi^2 = 0.89462      Levenberg-Marquardt = 0.001
 Free parameter values:
        3.913         0.6121          23.47         0.4709
------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 17.8706s    Reduced Chi^2 = 0.90125      Levenberg-Marquardt = 0.0001
 Free parameter values:
         3.78         0.6091          23.51         0.4717
 Total time = 18.6069s    Reduced Chi^2 = 0.88628      Levenberg-Marquardt = 0.001
 Free parameter values:
        3.781         0.6091          23.51         0.4717
------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 21.5306s    Reduced Chi^2 = 0.88679      Levenberg-Marquardt = 0.0001
 Free parameter values:
        4.003         0.6178          23.56         0.4711
 Total time = 22.2599s    Reduced Chi^2 = 0.90059      Levenberg-Marquardt = 0.001
 Free parameter values:
        4.002         0.6177          23.56         0.4711
 Total time = 23.0249s    Reduced Chi^2 = 0.87955      Levenberg-Marquardt = 0.01
 Free parameter values:
        3.993         0.6175          23.56         0.4711
------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 26.1153s    Reduced Chi^2 = 0.92951      Levenberg-Marquardt = 0.001
 Free parameter values:
        3.454         0.6017           23.5         0.4722
 Total time = 26.9197s    Reduced Chi^2 = 0.89961      Levenberg-Marquardt = 0.01
 Free parameter values:
        3.474         0.6022           23.5         0.4721
 Total time = 27.6846s    Reduced Chi^2 = 0.90234      Levenberg-Marquardt = 0.1
 Free parameter values:
        3.602         0.6055           23.5         0.4717
 Total time = 28.4854s    Reduced Chi^2 = 0.89403      Levenberg-Marquardt = 10
 Free parameter values:
        3.943         0.6153          23.54          0.471
 Total time = 29.2218s    Reduced Chi^2 = 0.88557      Levenberg-Marquardt = 1000
 Free parameter values:
        3.988         0.6173          23.56         0.4711
 Total time = 29.9737s    Reduced Chi^2 = 0.90452      Levenberg-Marquardt = 100000
 Free parameter values:
        3.993         0.6175          23.56         0.4711
 *** No improvement in chi-squared over previous iteration ***
 Total time = 29.9821s    Reduced Chi^2 = 0.87955
 Free parameter values:
        3.993         0.6175          23.56         0.4711
------------------------------------------------------------------------------
Fit converged:

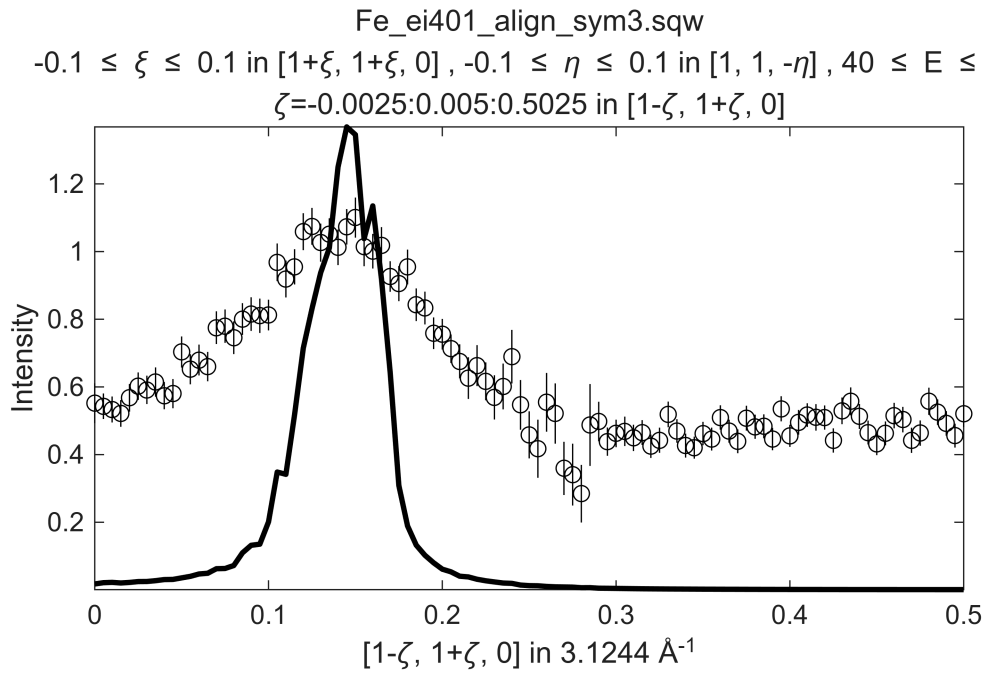
Parameter values (free parameters with error estimates):
---------------------------------------------------------
```

```
Foreground parameter values:
---------------------------
    1              1
    2              8
    3          3.993   +/-   0.09599
    4         0.6175   +/-   0.01321
    5              0
    6          23.56   +/-   0.307
    7              0
    8              0
    9              0
   10              0


Background parameter values:
---------------------------
    1         0.4711   +/-   0.004836
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000   -0.7422   -0.6754    0.0620
   -0.7422    1.0000    0.1549   -0.5420
   -0.6754    0.1549    1.0000   -0.0119
    0.0620   -0.5420   -0.0119    1.0000
```






```
*****************************
**** En = 55±5 going to 165
*****************************
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.88817s    Reduced Chi^2 = 16.3787
 Free parameter values:
         3.993         0.6175          23.56         0.5208
--------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 4.2549s    Reduced Chi^2 = 1.096      Levenberg-Marquardt = 0.1
 Free parameter values:
         7.055         0.6099          23.74         0.3889
--------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 7.5517s    Reduced Chi^2 = 1.0096      Levenberg-Marquardt = 0.01
 Free parameter values:
         10.12         0.6781          23.92          0.374
--------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 10.9364s    Reduced Chi^2 = 1.0148      Levenberg-Marquardt = 0.001
 Free parameter values:
         10.26         0.6836          23.85         0.3735
 Total time = 11.8196s    Reduced Chi^2 = 1.0136      Levenberg-Marquardt = 0.01
 Free parameter values:
         10.26         0.6834          23.85         0.3735
 Total time = 12.7635s    Reduced Chi^2 = 1.0248      Levenberg-Marquardt = 0.1
 Free parameter values:
         10.21         0.6822          23.85         0.3737
 Total time = 13.6848s    Reduced Chi^2 = 1.0063      Levenberg-Marquardt = 10
 Free parameter values:
         10.11         0.6791          23.89          0.374
--------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 17.0848s    Reduced Chi^2 = 1.0077      Levenberg-Marquardt = 1
 Free parameter values:
         9.985         0.6792          23.89         0.3741
 Total time = 17.9864s    Reduced Chi^2 = 1.0143      Levenberg-Marquardt = 10
 Free parameter values:
```

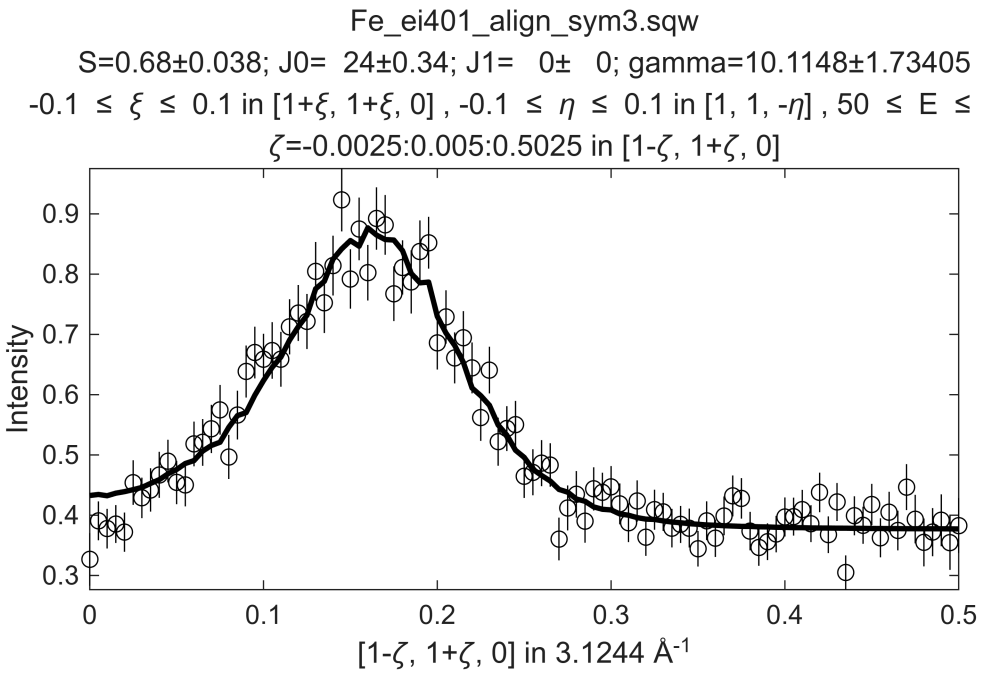
```
        10.06         0.6793          23.89          0.374
 Total time = 18.8517s    Reduced Chi^2 = 1.0189      Levenberg-Marquardt = 100
 Free parameter values:
         10.1         0.6792          23.89          0.374
 Total time = 19.6907s    Reduced Chi^2 = 1.0157      Levenberg-Marquardt = 10000
 Free parameter values:
        10.11         0.6791          23.89          0.374
 Total time = 20.5136s    Reduced Chi^2 = 1.0193      Levenberg-Marquardt = 1000000
 Free parameter values:
        10.11         0.6791          23.89          0.374
 Total time = 21.3032s    Reduced Chi^2 = 1.002      Levenberg-Marquardt = 100000000
 Free parameter values:
        10.11         0.6791          23.89          0.374
--------------------------------------------------------------------------------
Iteration = 5
-----------------
 Total time = 24.5374s    Reduced Chi^2 = 1.0155      Levenberg-Marquardt = 10000000
 Free parameter values:
        10.11         0.6791          23.89          0.374
 Total time = 25.355s    Reduced Chi^2 = 1.0288      Levenberg-Marquardt = 100000000
 Free parameter values:
        10.11         0.6791          23.89          0.374
 Total time = 26.2061s    Reduced Chi^2 = 1.0011      Levenberg-Marquardt = 1000000000
 Free parameter values:
        10.11         0.6791          23.89          0.374
--------------------------------------------------------------------------------
Fit converged:

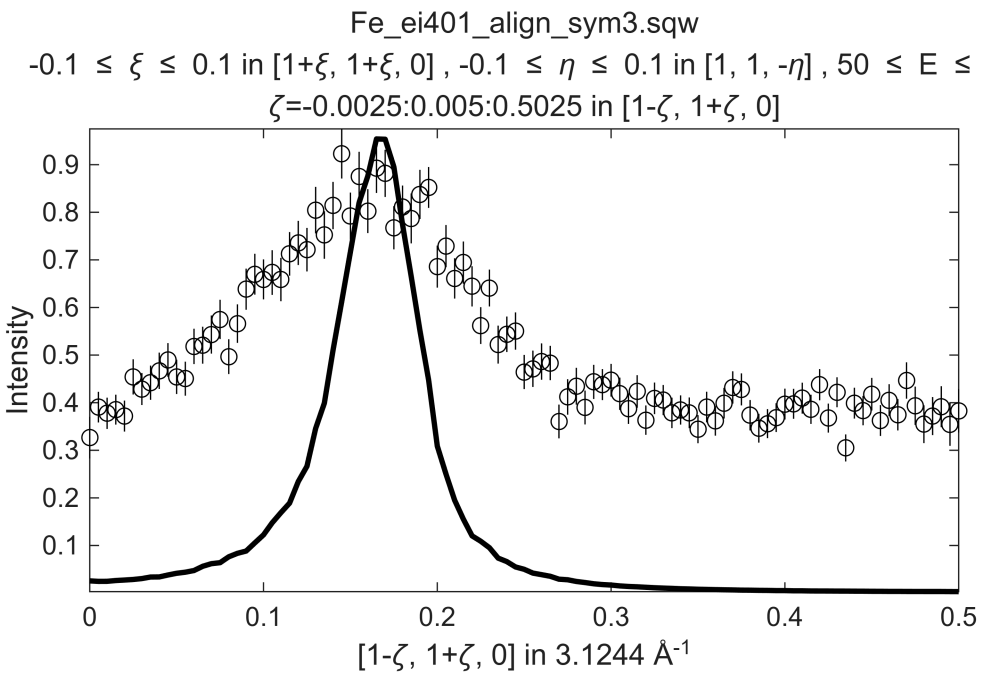
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          10.11   +/-   1.734
    4         0.6791   +/-   0.03808
    5              0
    6          23.89   +/-   0.3357
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1          0.374   +/-   0.006015
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.8697    0.2094   -0.6548
    0.8697    1.0000    0.2852   -0.7781
    0.2094    0.2852    1.0000   -0.0821
   -0.6548   -0.7781   -0.0821    1.0000
```






```
*****************************
**** En = 65±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
-----------------
 Total time = 0.73664s    Reduced Chi^2 = 18.992
 Free parameter values:
         10.11         0.6791          23.89         0.4679
-------------------------------------------------------------------------------
Iteration = 1
-----------------
 Total time = 3.7515s    Reduced Chi^2 = 1.0444      Levenberg-Marquardt = 0.1
 Free parameter values:
         12.77         0.6422          24.78         0.3324
-------------------------------------------------------------------------------
Iteration = 2
-----------------
 Total time = 6.7305s    Reduced Chi^2 = 0.94179      Levenberg-Marquardt = 0.01
 Free parameter values:
```

```
        18.23         0.7558          25.31         0.3114
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 10.6244s    Reduced Chi^2 = 0.94281      Levenberg-Marquardt = 0.001
 Free parameter values:
        19.25         0.7846          25.49         0.3091
 Total time = 11.7515s    Reduced Chi^2 = 0.94458      Levenberg-Marquardt = 0.01
 Free parameter values:
        19.12          0.782          25.48         0.3094
 Total time = 13.3808s    Reduced Chi^2 = 0.94618      Levenberg-Marquardt = 0.1
 Free parameter values:
        18.68         0.7723          25.43         0.3105
 Total time = 15.2444s    Reduced Chi^2 = 0.94046      Levenberg-Marquardt = 10
 Free parameter values:
        18.23         0.7587          25.33         0.3114
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 21.6056s    Reduced Chi^2 = 0.92903      Levenberg-Marquardt = 1
 Free parameter values:
        18.31         0.7645          25.37         0.3112
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 28.2916s    Reduced Chi^2 = 0.94032      Levenberg-Marquardt = 0.1
 Free parameter values:
        18.83         0.7746          25.44         0.3104
 Total time = 29.8966s    Reduced Chi^2 = 0.93685      Levenberg-Marquardt = 1
 Free parameter values:
        18.51         0.7682           25.4          0.311
 Total time = 31.4635s    Reduced Chi^2 = 0.95298      Levenberg-Marquardt = 10
 Free parameter values:
        18.38         0.7657          25.38         0.3112
 Total time = 33.0393s    Reduced Chi^2 = 0.9411      Levenberg-Marquardt = 1000
 Free parameter values:
        18.32         0.7647          25.37         0.3112
 Total time = 34.7443s    Reduced Chi^2 = 0.93899      Levenberg-Marquardt = 100000
 Free parameter values:
        18.31         0.7645          25.37         0.3112
 Total time = 36.1758s    Reduced Chi^2 = 0.93682      Levenberg-Marquardt = 10000000
 Free parameter values:
        18.31         0.7645          25.37         0.3112
 *** No improvement in chi-squared over previous iteration ***
 Total time = 36.1912s    Reduced Chi^2 = 0.92903
 Free parameter values:
        18.31         0.7645          25.37         0.3112
-------------------------------------------------------------------------------
Fit converged:

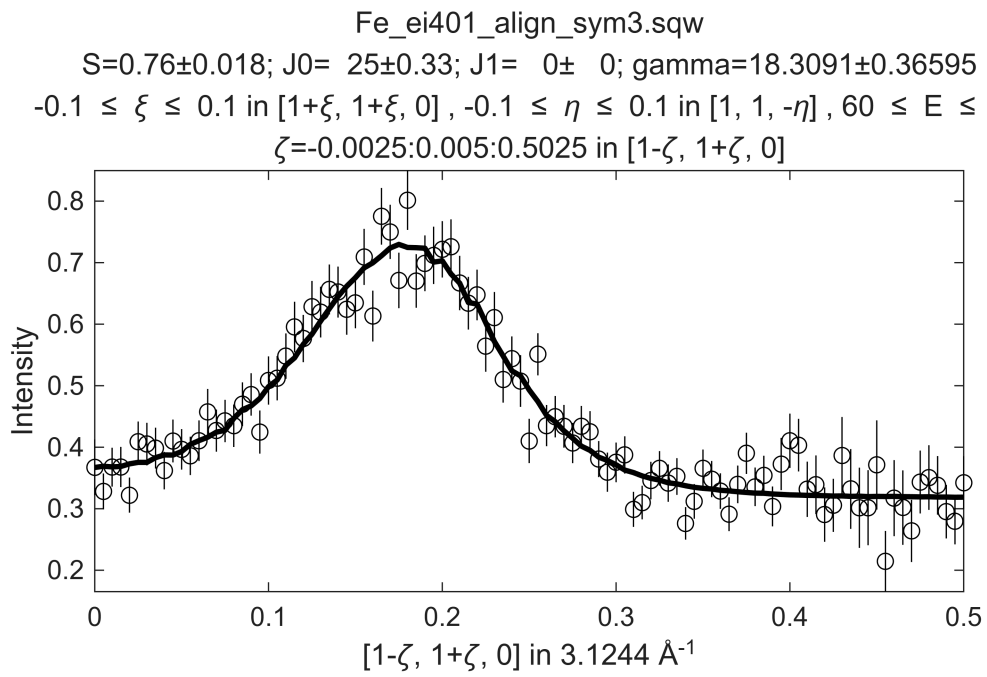
Parameter values (free parameters with error estimates):
---------------------------------------------------------

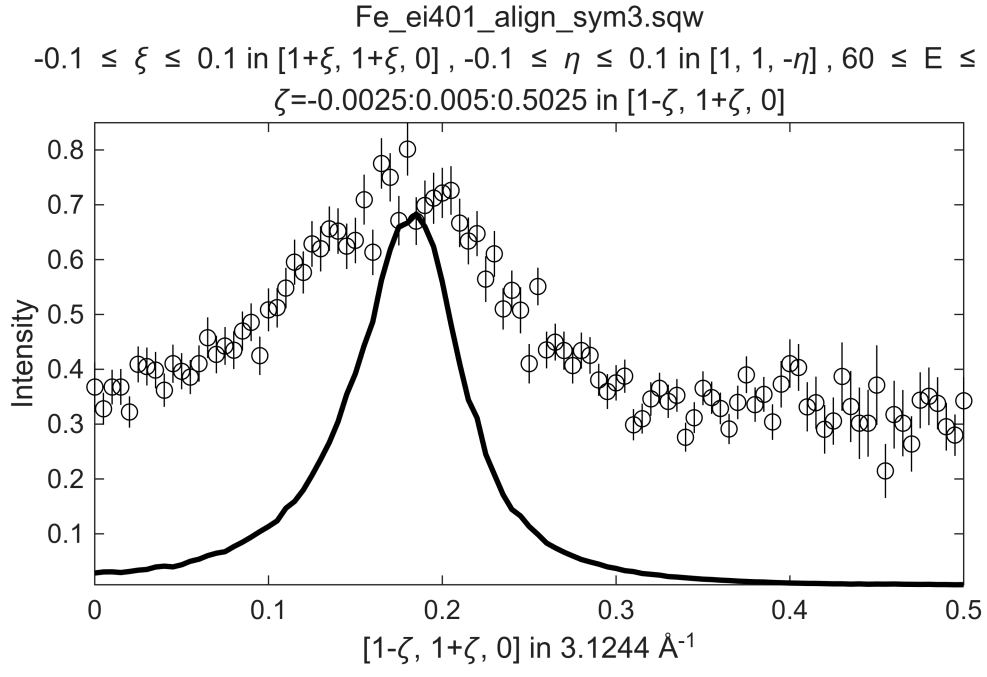
Foreground parameter values:
----------------------------
    1              1
    2              8
    3          18.31   +/-   0.3659
    4         0.7645   +/-   0.01818
    5              0
    6          25.37   +/-   0.3258
    7              0
    8              0
    9              0
   10              0


Background parameter values:
```

```
----------------------------
    1         0.3112   +/-   0.004967
    2              0
    3              0


Covariance matrix for free parameters:
-------------------------------------
    1.0000   -0.4663   -0.5873   -0.0858
   -0.4663    1.0000   -0.2133   -0.7344
   -0.5873   -0.2133    1.0000    0.2505
   -0.0858   -0.7344    0.2505    1.0000
```






```
*****************************
**** En = 75±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 1.7942s    Reduced Chi^2 = 23.7342
 Free parameter values:
```

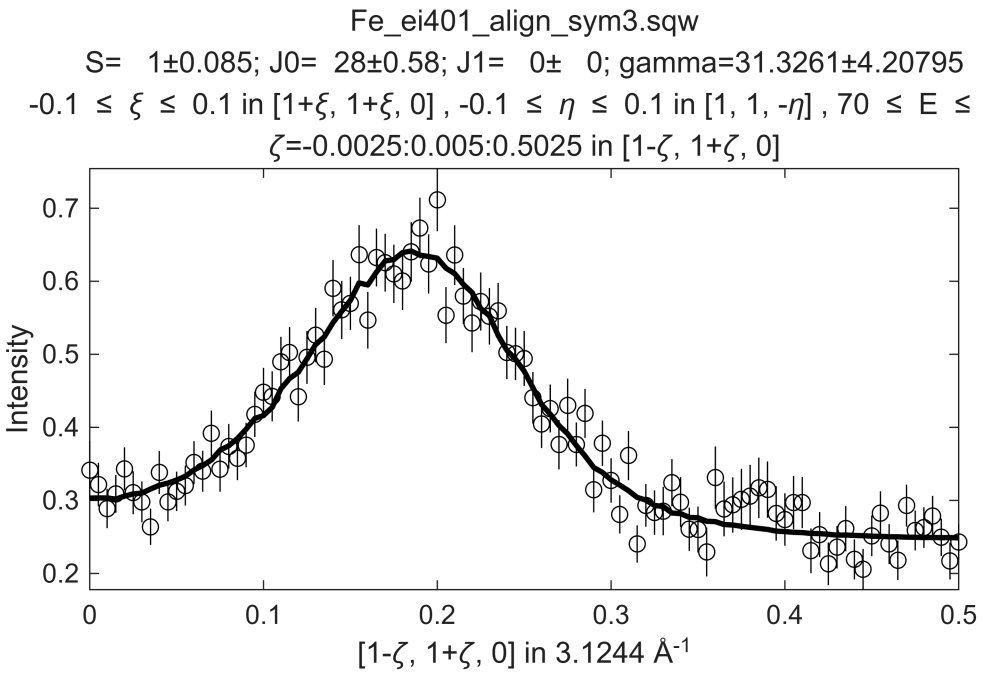
```
        18.31         0.7645          25.37         0.4008
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 8.5014s    Reduced Chi^2 = 1.0123      Levenberg-Marquardt = 0.1
 Free parameter values:
        21.31         0.7791          26.88         0.2611
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 15.6534s    Reduced Chi^2 = 0.87636      Levenberg-Marquardt = 0.01
 Free parameter values:
        29.01         0.9512          27.74         0.2377
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 22.3758s    Reduced Chi^2 = 0.85411      Levenberg-Marquardt = 0.001
 Free parameter values:
        31.56          1.014          28.12         0.2343
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 28.64s    Reduced Chi^2 = 0.85963      Levenberg-Marquardt = 0.0001
 Free parameter values:
        31.29           1.01          28.09         0.2346
 Total time = 30.3457s    Reduced Chi^2 = 0.85974      Levenberg-Marquardt = 0.001
 Free parameter values:
        31.29           1.01          28.09         0.2346
 Total time = 32.1993s    Reduced Chi^2 = 0.8541      Levenberg-Marquardt = 0.01
 Free parameter values:
        31.33           1.01          28.09         0.2346
-------------------------------------------------------------------------------
Fit converged:

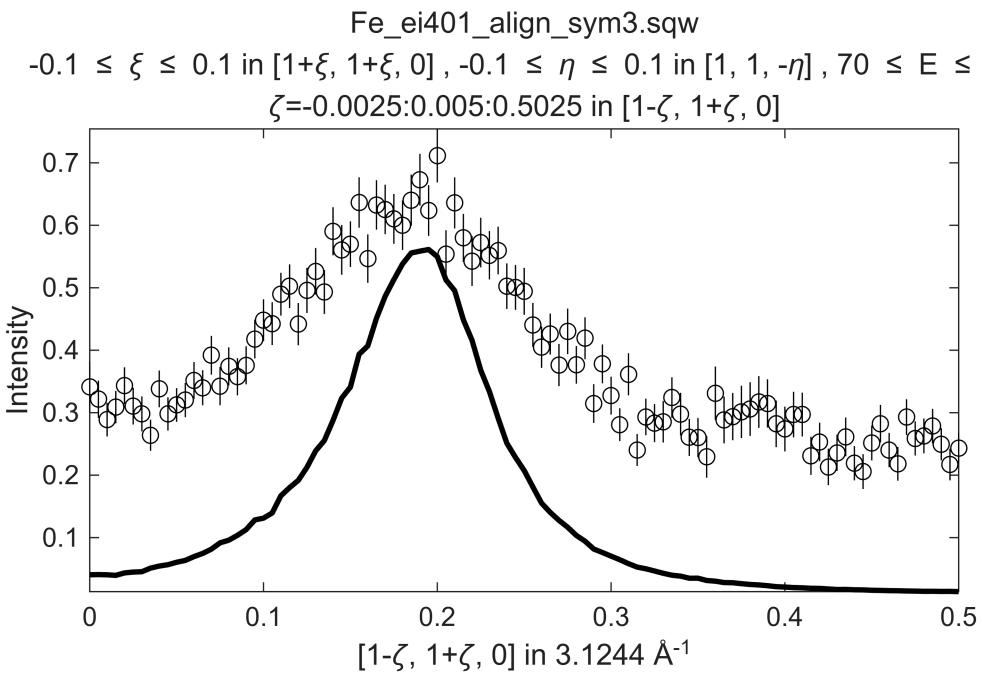
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          31.33   +/-   4.208
    4           1.01   +/-   0.08528
    5              0
    6          28.09   +/-   0.58
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1         0.2346   +/-   0.007022
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9534    0.8138   -0.7874
    0.9534    1.0000    0.7865   -0.8811
    0.8138    0.7865    1.0000   -0.5844
   -0.7874   -0.8811   -0.5844    1.0000
```






```
*****************************
**** En = 85±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 2.052s    Reduced Chi^2 = 30.8691
 Free parameter values:
         31.33           1.01          28.09          0.329
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 10.2216s    Reduced Chi^2 = 0.81857      Levenberg-Marquardt = 0.1
 Free parameter values:
         29.86         0.9059          28.61         0.1978
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 17.7964s    Reduced Chi^2 = 0.70229      Levenberg-Marquardt = 0.01
 Free parameter values:
```

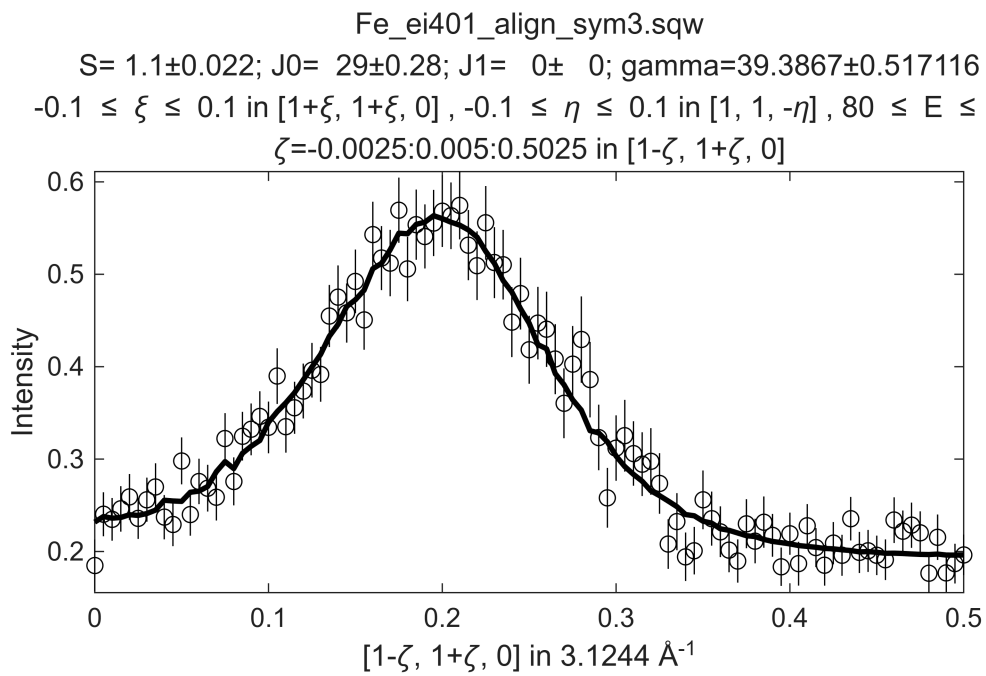
```
         37.31          1.069          29.12         0.1786
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 26.3224s    Reduced Chi^2 = 0.69006      Levenberg-Marquardt = 0.001
 Free parameter values:
          39.4          1.118          29.42          0.176
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 33.8892s    Reduced Chi^2 = 0.69166      Levenberg-Marquardt = 0.0001
 Free parameter values:
         38.89          1.109           29.4         0.1767
 Total time = 35.7857s    Reduced Chi^2 = 0.69783      Levenberg-Marquardt = 0.001
 Free parameter values:
          38.9           1.11           29.4         0.1766
 Total time = 37.5561s    Reduced Chi^2 = 0.69833      Levenberg-Marquardt = 0.01
 Free parameter values:
         38.98          1.111          29.41         0.1765
 Total time = 39.4932s    Reduced Chi^2 = 0.69841      Levenberg-Marquardt = 1
 Free parameter values:
         39.29          1.117          29.43         0.1762
 Total time = 41.549s     Reduced Chi^2 = 0.6833      Levenberg-Marquardt = 100
 Free parameter values:
         39.39          1.118          29.42         0.1761
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 49.6338s    Reduced Chi^2 = 0.69151      Levenberg-Marquardt = 10
 Free parameter values:
         39.34          1.117          29.41         0.1761
 Total time = 51.9622s    Reduced Chi^2 = 0.69975      Levenberg-Marquardt = 100
 Free parameter values:
         39.37          1.118          29.42         0.1761
 Total time = 54.0483s    Reduced Chi^2 = 0.68782      Levenberg-Marquardt = 1000
 Free parameter values:
         39.38          1.118          29.42         0.1761
 Total time = 56.0766s    Reduced Chi^2 = 0.69637      Levenberg-Marquardt = 100000
 Free parameter values:
         39.39          1.118          29.42         0.1761
 Total time = 58.3317s    Reduced Chi^2 = 0.69615      Levenberg-Marquardt = 10000000
 Free parameter values:
         39.39          1.118          29.42         0.1761
 Total time = 60.3888s    Reduced Chi^2 = 0.6922      Levenberg-Marquardt = 1000000000
 Free parameter values:
         39.39          1.118          29.42         0.1761
 *** No improvement in chi-squared over previous iteration ***
 Total time = 60.4309s    Reduced Chi^2 = 0.6833
 Free parameter values:
         39.39          1.118          29.42         0.1761
-------------------------------------------------------------------------------
Fit converged:

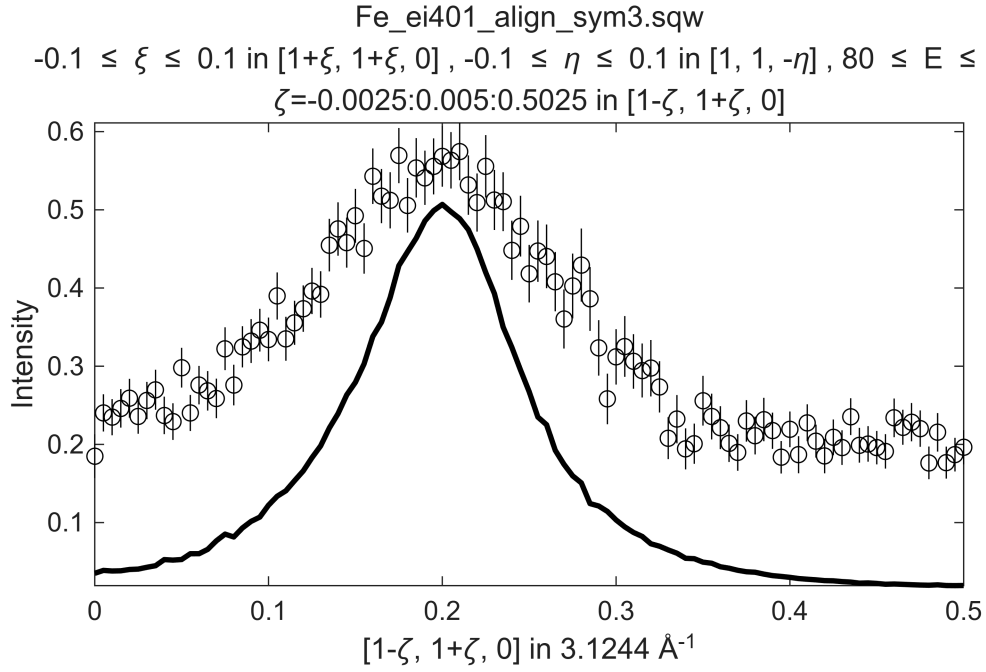
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          39.39   +/-   0.5171
    4          1.118   +/-   0.02212
    5              0
    6          29.42   +/-   0.2754
    7              0
    8              0
    9              0
   10              0
```

```
Background parameter values:
----------------------------
    1         0.1761   +/-   0.003509
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.0325   -0.5431   -0.4350
    0.0325    1.0000   -0.3723   -0.8400
   -0.5431   -0.3723    1.0000    0.2572
   -0.4350   -0.8400    0.2572    1.0000
```






```
****************************
**** En = 95±5 going to 165
****************************
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------
Starting point
```

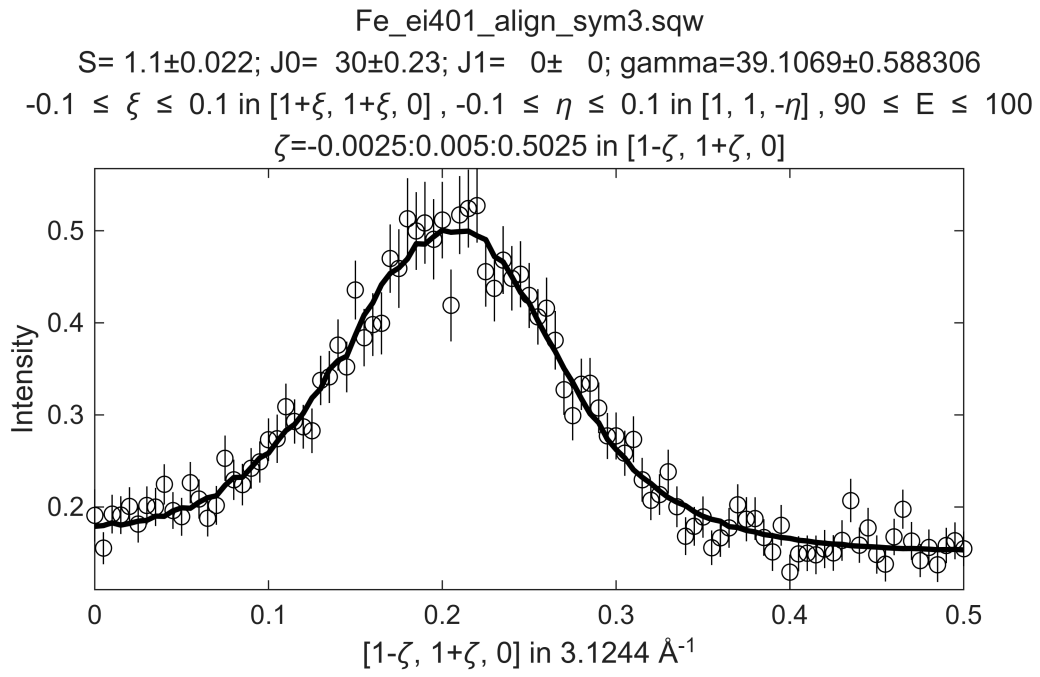
```
------------------
 Total time = 1.9203s    Reduced Chi^2 = 35.9552
 Free parameter values:
        39.39          1.118          29.42          0.2635
--------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 10.7735s    Reduced Chi^2 = 0.82269      Levenberg-Marquardt = 0.1
 Free parameter values:
        32.43          0.904          29.91          0.1513
--------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 19.5381s    Reduced Chi^2 = 0.6995      Levenberg-Marquardt = 0.01
 Free parameter values:
        38.39          1.042          30.31          0.1346
--------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 27.9085s    Reduced Chi^2 = 0.69456      Levenberg-Marquardt = 0.001
 Free parameter values:
        39.16          1.065          30.45          0.1335
--------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 36.7343s    Reduced Chi^2 = 0.69958      Levenberg-Marquardt = 0.0001
 Free parameter values:
        39.36           1.07          30.47          0.1332
 Total time = 39.0755s    Reduced Chi^2 = 0.69821      Levenberg-Marquardt = 0.001
 Free parameter values:
        39.36          1.069          30.47          0.1332
 Total time = 41.4158s    Reduced Chi^2 = 0.69238      Levenberg-Marquardt = 0.01
 Free parameter values:
        39.33          1.069          30.47          0.1333
--------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 50.4578s    Reduced Chi^2 = 0.69093      Levenberg-Marquardt = 0.001
 Free parameter values:
        39.11          1.065          30.45          0.1336
--------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 59.8348s    Reduced Chi^2 = 0.69284      Levenberg-Marquardt = 0.0001
 Free parameter values:
        39.34          1.068          30.48          0.1333
 Total time = 62.1884s    Reduced Chi^2 = 0.6959      Levenberg-Marquardt = 0.001
 Free parameter values:
        39.34          1.068          30.48          0.1333
 Total time = 64.435s    Reduced Chi^2 = 0.69553      Levenberg-Marquardt = 0.01
 Free parameter values:
        39.31          1.068          30.47          0.1333
 Total time = 66.777s    Reduced Chi^2 = 0.69389      Levenberg-Marquardt = 1
 Free parameter values:
        39.16          1.065          30.46          0.1335
 Total time = 69.0321s    Reduced Chi^2 = 0.69241      Levenberg-Marquardt = 100
 Free parameter values:
        39.11          1.065          30.45          0.1336
 Total time = 71.3301s    Reduced Chi^2 = 0.69919      Levenberg-Marquardt = 10000
 Free parameter values:
        39.11          1.065          30.45          0.1336
 *** No improvement in chi-squared over previous iteration ***
 Total time = 71.3492s    Reduced Chi^2 = 0.69093
 Free parameter values:
        39.11          1.065          30.45          0.1336
--------------------------------------------------------------------------------
Fit converged:
```

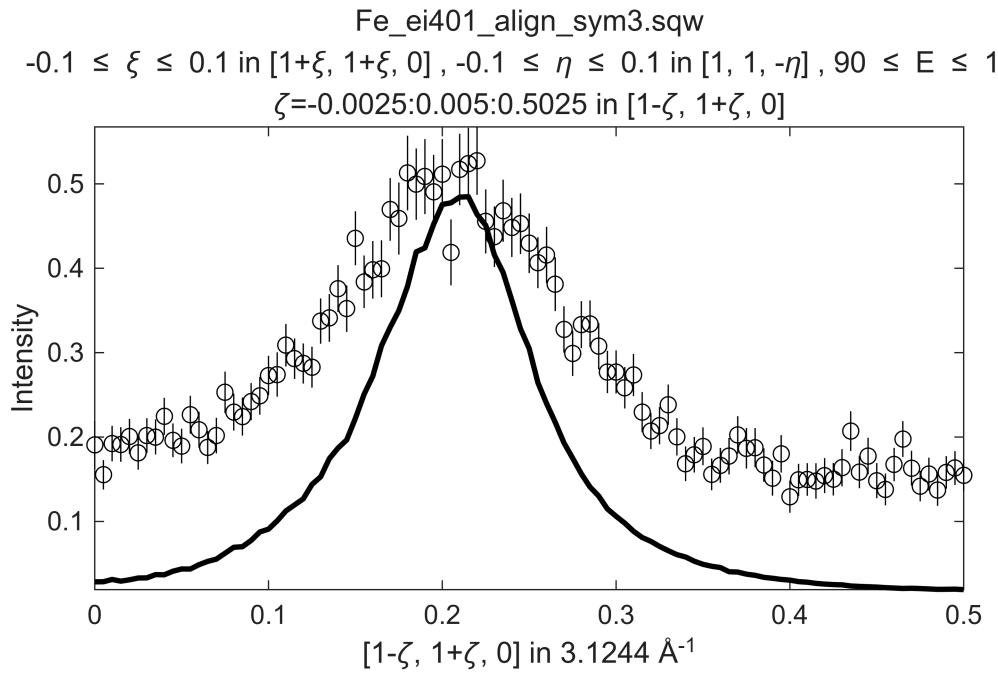
```
Parameter values (free parameters with error estimates):
-------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          39.11   +/-   0.5883
    4          1.065   +/-   0.02211
    5              0
    6          30.45   +/-   0.228
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1         0.1336   +/-   0.003165
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.0595   -0.4884   -0.5315
    0.0595    1.0000   -0.2552   -0.8345
   -0.4884   -0.2552    1.0000    0.2180
   -0.5315   -0.8345    0.2180    1.0000
```






```
*****************************
**** En = 105±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 3.2989s    Reduced Chi^2 = 52.2814
 Free parameter values:
         39.11          1.065          30.45         0.2347
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 17.944s    Reduced Chi^2 = 1.2457      Levenberg-Marquardt = 0.1
 Free parameter values:
         36.52         0.9267          31.16         0.1135
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 31.8598s    Reduced Chi^2 = 1.0381      Levenberg-Marquardt = 0.01
 Free parameter values:
            46          1.118          31.89        0.09319
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 45.5813s    Reduced Chi^2 = 1.0262      Levenberg-Marquardt = 0.001
 Free parameter values:
         49.05          1.183          32.29        0.08998
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 59.9394s    Reduced Chi^2 = 1.0222      Levenberg-Marquardt = 0.0001
 Free parameter values:
         49.09          1.185          32.29        0.08996
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 73.6349s    Reduced Chi^2 = 1.0294      Levenberg-Marquardt = 1e-05
 Free parameter values:
            49          1.182          32.28        0.09004
 Total time = 77.098s    Reduced Chi^2 = 1.0255      Levenberg-Marquardt = 0.0001
 Free parameter values:
            49          1.182          32.28        0.09004
 Total time = 80.9403s    Reduced Chi^2 = 1.0306      Levenberg-Marquardt = 0.001
 Free parameter values:
```

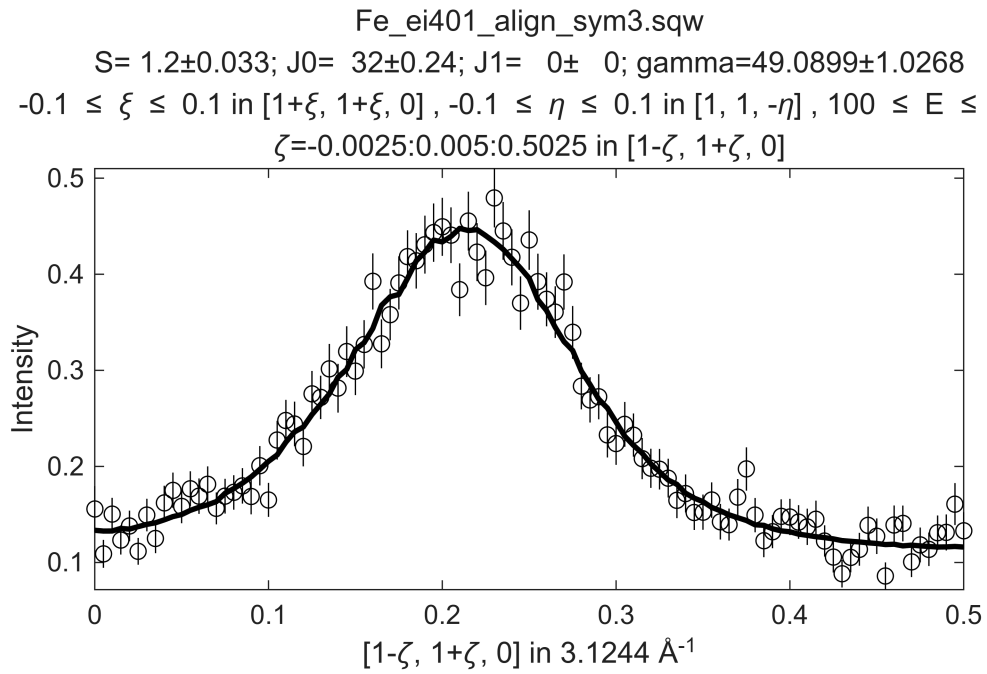
```
        49.01          1.182          32.28        0.09004
 Total time = 84.3608s    Reduced Chi^2 = 1.0312      Levenberg-Marquardt = 0.1
 Free parameter values:
        49.07          1.183          32.29        0.08997
 Total time = 87.9313s    Reduced Chi^2 = 1.0296      Levenberg-Marquardt = 10
 Free parameter values:
         49.1          1.184          32.29        0.08995
 Total time = 91.1265s    Reduced Chi^2 = 1.032      Levenberg-Marquardt = 1000
 Free parameter values:
        49.09          1.185          32.29        0.08996
 *** No improvement in chi-squared over previous iteration ***
 Total time = 91.1511s    Reduced Chi^2 = 1.0222
 Free parameter values:
        49.09          1.185          32.29        0.08996
--------------------------------------------------------------------------------
Fit converged:

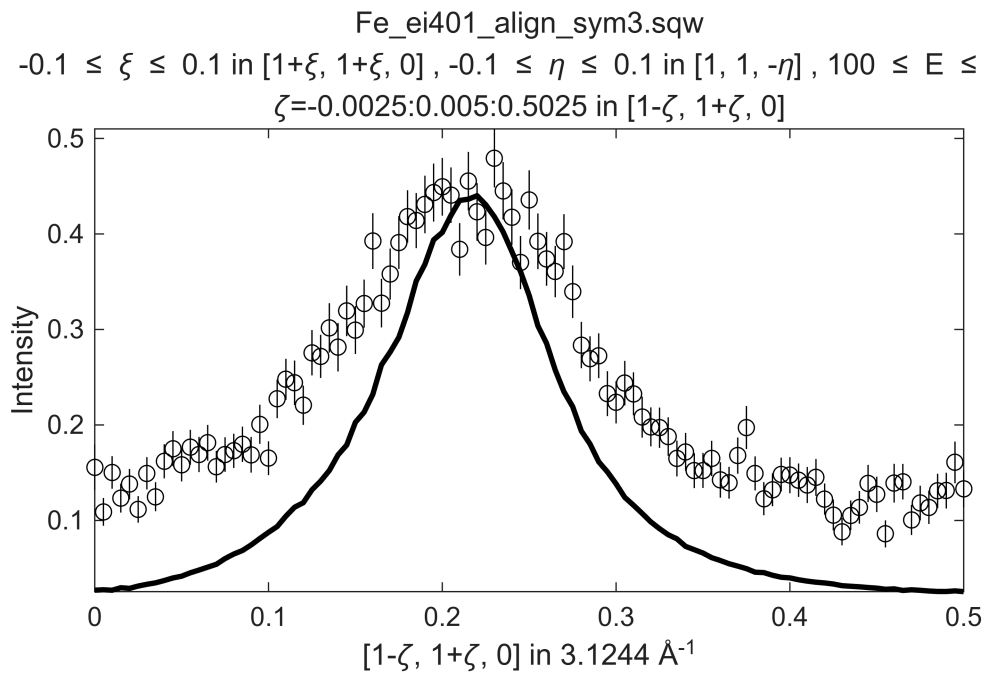
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          49.09   +/-   1.027
    4          1.185   +/-   0.03273
    5              0
    6          32.29   +/-   0.2425
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.08996   +/-   0.003799
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.5648   -0.2879   -0.8143
    0.5648    1.0000   -0.0865   -0.9181
   -0.2879   -0.0865    1.0000    0.0221
   -0.8143   -0.9181    0.0221    1.0000
```






```
*****************************
**** En = 115±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 3.6259s    Reduced Chi^2 = 66.8149
 Free parameter values:
         49.09          1.185          32.29         0.2005
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 18.645s    Reduced Chi^2 = 0.82697      Levenberg-Marquardt = 0.1
 Free parameter values:
         41.16         0.8831          32.91        0.09279
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 33.3461s    Reduced Chi^2 = 0.6326      Levenberg-Marquardt = 0.01
 Free parameter values:
```

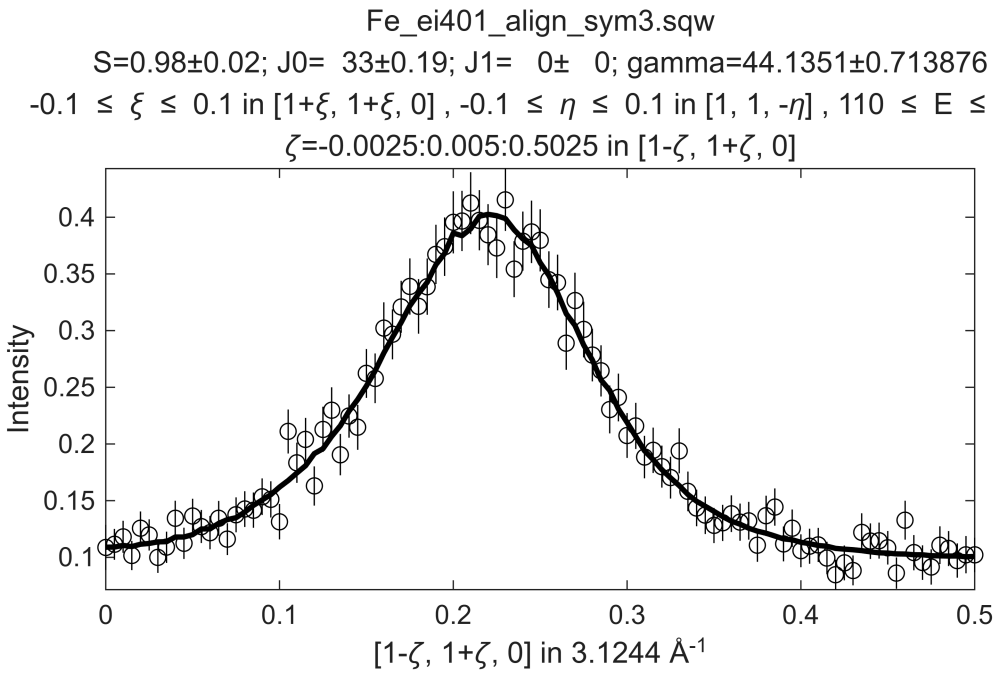
```
        44.14         0.9818          33.18        0.08017
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 48.3479s    Reduced Chi^2 = 0.64565      Levenberg-Marquardt = 0.001
 Free parameter values:
        45.21          1.003          33.29        0.07884
 Total time = 52.4022s    Reduced Chi^2 = 0.6366      Levenberg-Marquardt = 0.01
 Free parameter values:
        45.05              1          33.28          0.079
 Total time = 56.2022s    Reduced Chi^2 = 0.63714      Levenberg-Marquardt = 0.1
 Free parameter values:
        44.57         0.9922          33.23        0.07952
 Total time = 59.964s    Reduced Chi^2 = 0.64102      Levenberg-Marquardt = 10
 Free parameter values:
        44.15         0.9832          33.18         0.0801
 Total time = 64.0128s    Reduced Chi^2 = 0.63273      Levenberg-Marquardt = 1000
 Free parameter values:
        44.14         0.9819          33.18        0.08017
 Total time = 67.629s    Reduced Chi^2 = 0.63801      Levenberg-Marquardt = 100000
 Free parameter values:
        44.14         0.9818          33.18        0.08017
 *** No improvement in chi-squared over previous iteration ***
 Total time = 67.6665s    Reduced Chi^2 = 0.6326
 Free parameter values:
        44.14         0.9818          33.18        0.08017
-------------------------------------------------------------------------------
Fit converged:

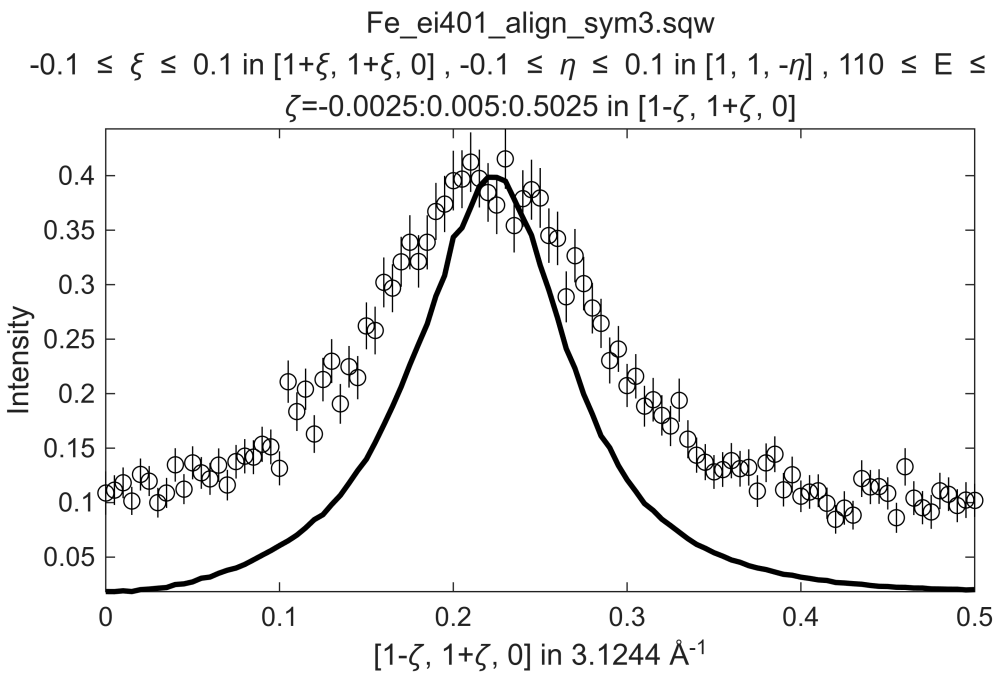
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          44.14   +/-   0.7139
    4         0.9818   +/-   0.01999
    5              0
    6          33.18   +/-   0.1918
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.08017   +/-   0.002468
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.4867   -0.3171   -0.7876
    0.4867    1.0000   -0.1044   -0.8984
   -0.3171   -0.1044    1.0000    0.0467
   -0.7876   -0.8984    0.0467    1.0000
```






```
*****************************
**** En = 125±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------
Starting point
------------------
 Total time = 4.1893s    Reduced Chi^2 = 71.132
 Free parameter values:
         44.14         0.9818          33.18         0.1772
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 20.5315s    Reduced Chi^2 = 1.0836      Levenberg-Marquardt = 0.1
 Free parameter values:
         36.58         0.8372          32.63        0.07173
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 37.488s    Reduced Chi^2 = 0.80326      Levenberg-Marquardt = 0.01
 Free parameter values:
```

```
        44.97         0.9899          33.01        0.05572
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 53.5921s    Reduced Chi^2 = 0.79727      Levenberg-Marquardt = 0.001
 Free parameter values:
        45.92          1.016           33.2        0.05436
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 70.236s    Reduced Chi^2 = 0.80373      Levenberg-Marquardt = 0.0001
 Free parameter values:
        46.23          1.022          33.22        0.05411
 Total time = 74.3136s    Reduced Chi^2 = 0.80695      Levenberg-Marquardt = 0.001
 Free parameter values:
        46.23          1.022          33.22        0.05411
 Total time = 78.6497s    Reduced Chi^2 = 0.80121      Levenberg-Marquardt = 0.01
 Free parameter values:
        46.19          1.021          33.22        0.05415
 Total time = 83.1832s    Reduced Chi^2 = 0.80126      Levenberg-Marquardt = 1
 Free parameter values:
        45.98          1.018           33.2        0.05436
 Total time = 87.4076s    Reduced Chi^2 = 0.79834      Levenberg-Marquardt = 100
 Free parameter values:
        45.93          1.016           33.2        0.05437
 Total time = 91.7649s    Reduced Chi^2 = 0.79406      Levenberg-Marquardt = 10000
 Free parameter values:
        45.92          1.016           33.2        0.05436
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 108.7256s    Reduced Chi^2 = 0.80184      Levenberg-Marquardt = 1000
 Free parameter values:
        45.93          1.016           33.2        0.05436
 Total time = 112.9252s    Reduced Chi^2 = 0.80186      Levenberg-Marquardt = 10000
 Free parameter values:
        45.93          1.016           33.2        0.05436
 Total time = 117.0809s    Reduced Chi^2 = 0.80619      Levenberg-Marquardt = 100000
 Free parameter values:
        45.92          1.016           33.2        0.05436
 Total time = 121.0452s    Reduced Chi^2 = 0.80183      Levenberg-Marquardt = 10000000
 Free parameter values:
        45.92          1.016           33.2        0.05436
 Total time = 125.0638s    Reduced Chi^2 = 0.81013      Levenberg-Marquardt = 1000000000
 Free parameter values:
        45.92          1.016           33.2        0.05436
 Total time = 129.1683s    Reduced Chi^2 = 0.79468      Levenberg-Marquardt = 100000000000
 Free parameter values:
        45.92          1.016           33.2        0.05436
 *** No improvement in chi-squared over previous iteration ***
 Total time = 129.1951s    Reduced Chi^2 = 0.79406
 Free parameter values:
        45.92          1.016           33.2        0.05436
-------------------------------------------------------------------------------
Fit converged:

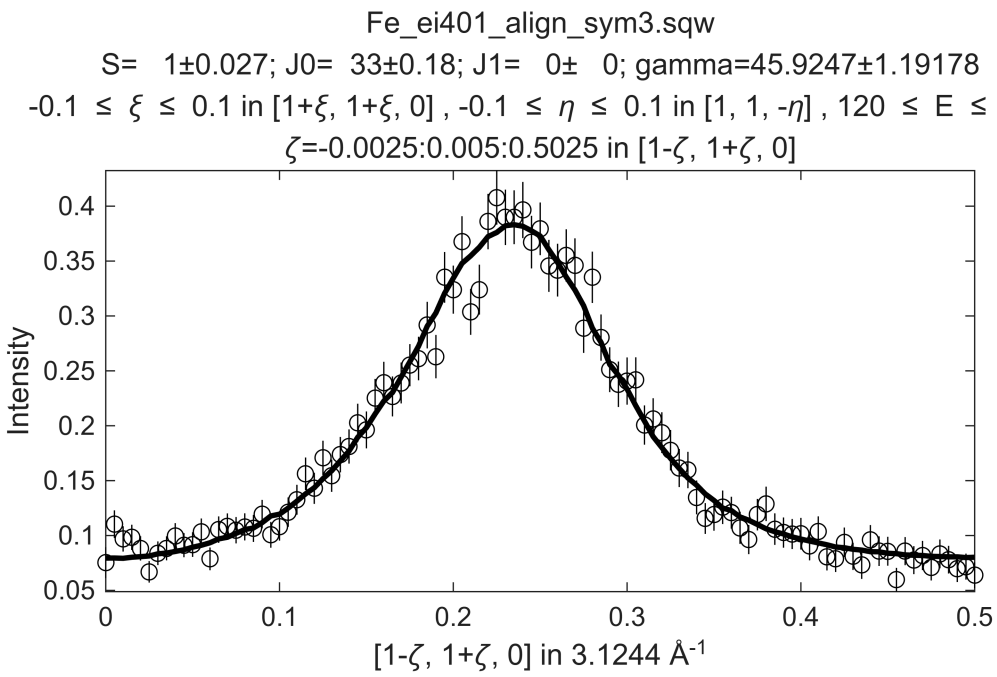
Parameter values (free parameters with error estimates):
---------------------------------------------------------

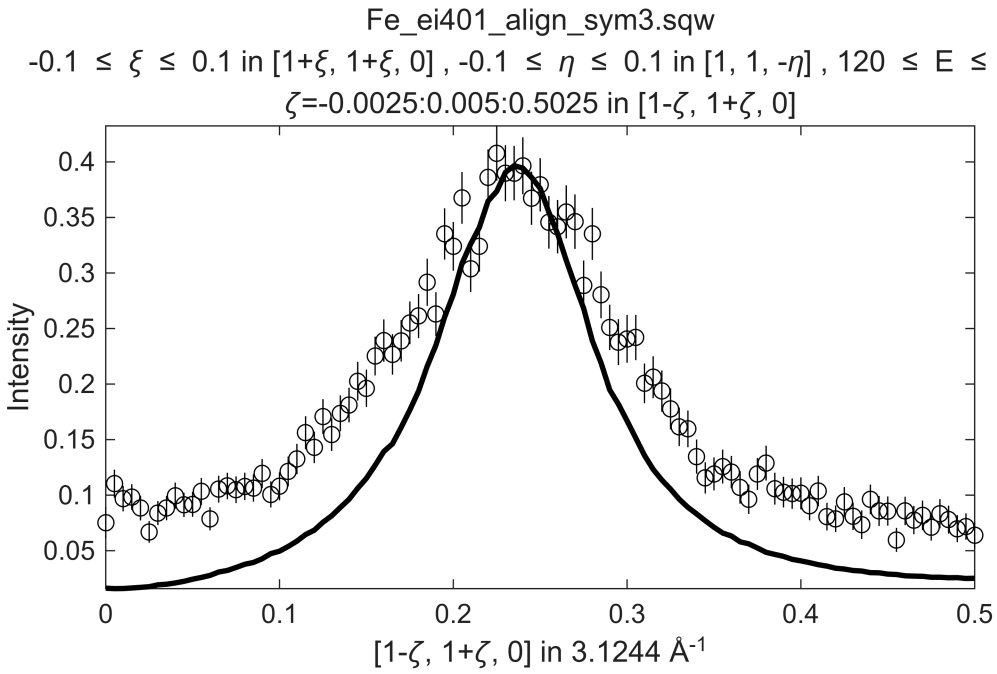
Foreground parameter values:
----------------------------
    1             1
    2             8
    3         45.92   +/-   1.192
    4         1.016   +/-   0.02724
    5             0
    6          33.2   +/-   0.1782
    7             0
```

```
     8              0
     9              0
    10              0


Background parameter values:
----------------------------
     1        0.05436   +/-   0.002751
     2              0
     3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.7625    0.0899   -0.9256
    0.7625    1.0000    0.1882   -0.9412
    0.0899    0.1882    1.0000   -0.2523
   -0.9256   -0.9412   -0.2523    1.0000
```






```
*****************************
**** En = 135±5 going to 165
*****************************
```

```
------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------
Starting point
------------------
 Total time = 3.9608s    Reduced Chi^2 = 96.4322
 Free parameter values:
        45.92          1.016           33.2         0.1601
------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 19.7367s    Reduced Chi^2 = 1.3825      Levenberg-Marquardt = 0.1
 Free parameter values:
        41.32         0.8476          33.43        0.05792
------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 35.2166s    Reduced Chi^2 = 1.0347      Levenberg-Marquardt = 0.01
 Free parameter values:
        49.54          0.996          33.97         0.0427
------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 52.8071s    Reduced Chi^2 = 1.0231      Levenberg-Marquardt = 0.001
 Free parameter values:
        50.89          1.026          34.13        0.04117
------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 71.9463s    Reduced Chi^2 = 1.0203      Levenberg-Marquardt = 0.0001
 Free parameter values:
        50.99          1.028          34.14        0.04106
------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 91.3706s    Reduced Chi^2 = 1.0279      Levenberg-Marquardt = 1e-05
 Free parameter values:
        50.88          1.026          34.12        0.04114
 Total time = 95.9168s    Reduced Chi^2 = 1.0247      Levenberg-Marquardt = 0.0001
 Free parameter values:
        50.88          1.026          34.12        0.04113
 Total time = 100.6904s    Reduced Chi^2 = 1.0327      Levenberg-Marquardt = 0.001
 Free parameter values:
        50.88          1.026          34.12        0.04113
 Total time = 105.3811s    Reduced Chi^2 = 1.0251      Levenberg-Marquardt = 0.1
 Free parameter values:
        50.94          1.027          34.12        0.04108
 Total time = 110.6658s    Reduced Chi^2 = 1.0321      Levenberg-Marquardt = 10
 Free parameter values:
        50.99          1.028          34.13        0.04105
 Total time = 116.0614s    Reduced Chi^2 = 1.0277      Levenberg-Marquardt = 1000
 Free parameter values:
        50.99          1.028          34.14        0.04106
 *** No improvement in chi-squared over previous iteration ***
 Total time = 116.0917s    Reduced Chi^2 = 1.0203
 Free parameter values:
        50.99          1.028          34.14        0.04106
------------------------------------------------------------------------------
Fit converged:

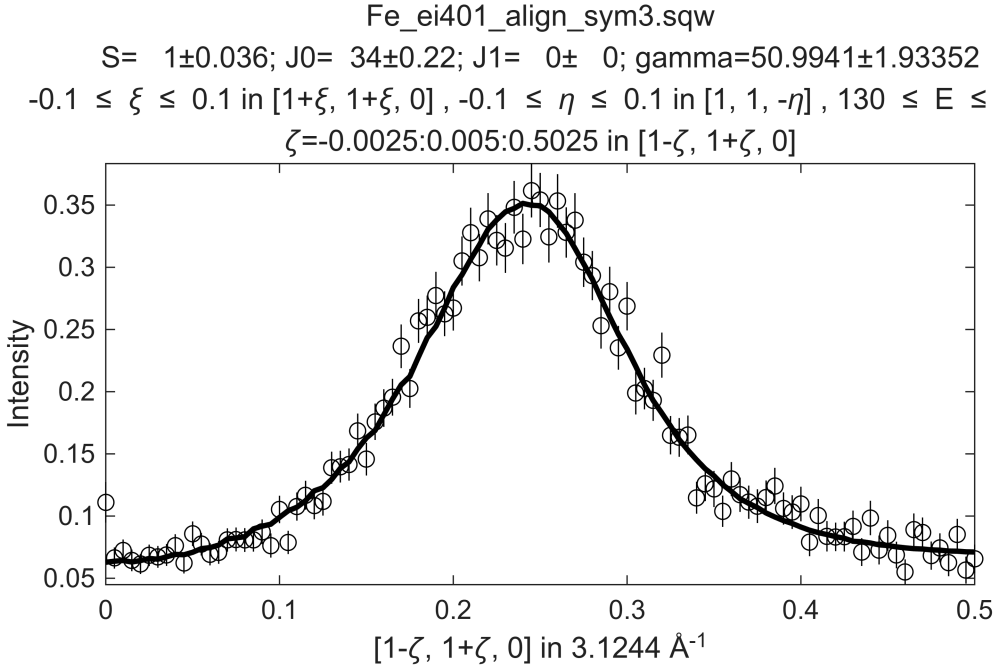
Parameter values (free parameters with error estimates):
---------------------------------------------------------

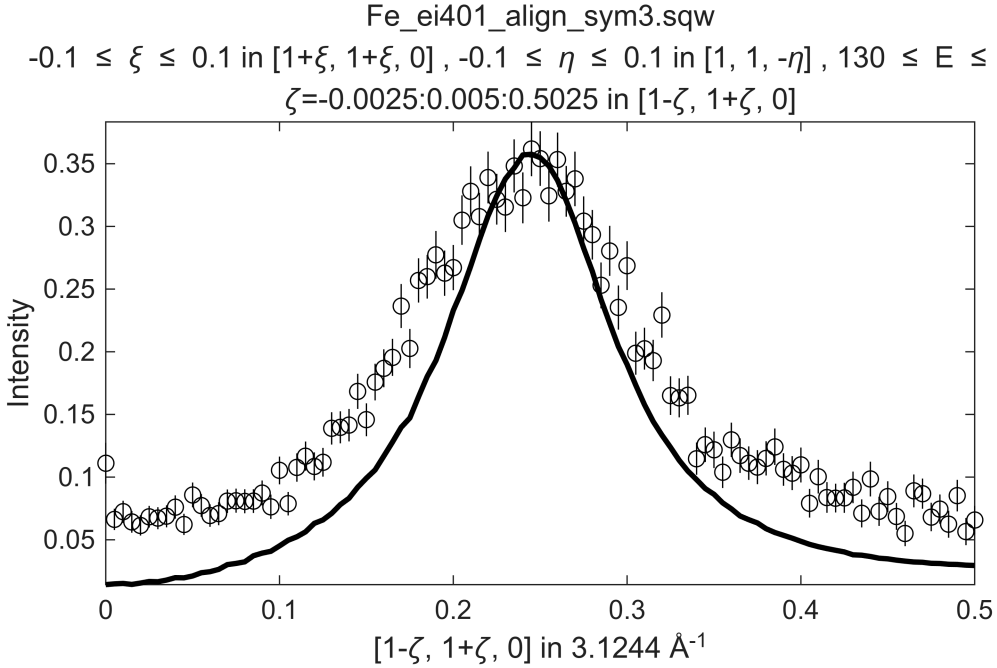
Foreground parameter values:
----------------------------
    1             1
    2             8
    3          50.99   +/-   1.934
    4          1.028   +/-   0.03646
```

```
     5              0
     6          34.14   +/-   0.216
     7              0
     8              0
     9              0
    10              0


Background parameter values:
----------------------------
     1        0.04106   +/-   0.003061
     2              0
     3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.8787    0.4582   -0.9664
    0.8787    1.0000    0.5187   -0.9670
    0.4582    0.5187    1.0000   -0.5642
   -0.9664   -0.9670   -0.5642    1.0000
```





```
****************************
**** En = 145±5 going to 165
****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 6.1699s    Reduced Chi^2 = 129.4171
 Free parameter values:
        50.99          1.028          34.14         0.1503
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 29.8743s    Reduced Chi^2 = 1.7325      Levenberg-Marquardt = 0.1
 Free parameter values:
        44.65         0.8633          34.34        0.04871
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 53.4144s    Reduced Chi^2 = 1.2613      Levenberg-Marquardt = 0.01
 Free parameter values:
        52.97           1.01          34.84        0.03422
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 77.6085s    Reduced Chi^2 = 1.2555      Levenberg-Marquardt = 0.001
 Free parameter values:
        54.09          1.036          35.01        0.03308
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 101.1064s    Reduced Chi^2 = 1.2479      Levenberg-Marquardt = 0.0001
 Free parameter values:
        54.19          1.037          35.02        0.03302
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 126.5198s    Reduced Chi^2 = 1.2504      Levenberg-Marquardt = 1e-05
 Free parameter values:
        54.12          1.037          35.01        0.03303
 Total time = 132.8087s    Reduced Chi^2 = 1.2557      Levenberg-Marquardt = 0.0001
 Free parameter values:
        54.12          1.037          35.01        0.03303
 Total time = 139.3282s    Reduced Chi^2 = 1.2473      Levenberg-Marquardt = 0.001
 Free parameter values:
        54.12          1.037          35.01        0.03303
-------------------------------------------------------------------------------
Fit converged:

Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          54.12   +/-   3.068
    4          1.037   +/-   0.04435
    5              0
    6          35.01   +/-   0.3204
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
```

```
    1        0.03303   +/-   0.002834
    2              0
    3              0

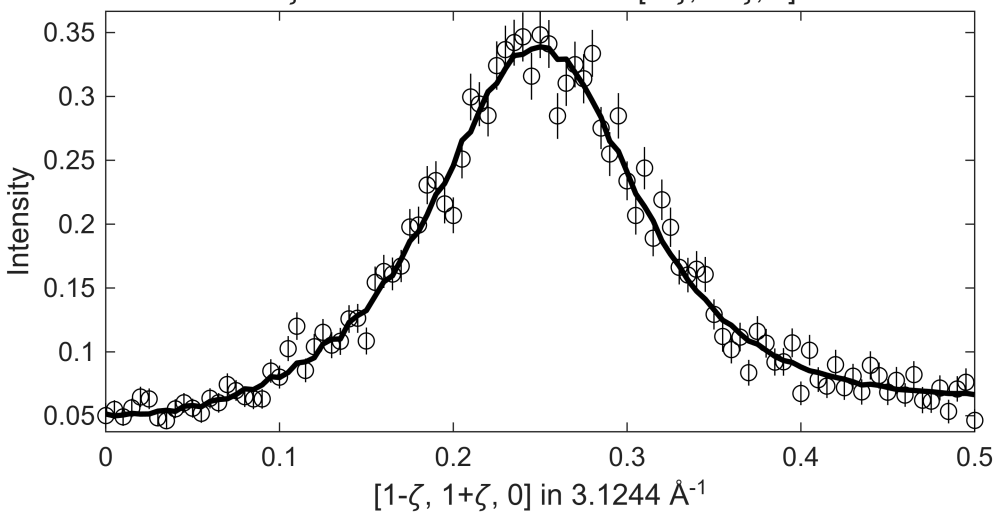

Covariance matrix for free parameters:
-------------------------------------
    1.0000    0.9139    0.7984   -0.7980
    0.9139    1.0000    0.7534   -0.8987
    0.7984    0.7534    1.0000   -0.5902
   -0.7980   -0.8987   -0.5902    1.0000
```

Fe_ei401_align_sym3.sqw

S= 1±0.044; J0= 35±0.32; J1= 0± 0; gamma=54.1168±3.06796

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1+ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1, -η] , 140 ≤ E ≤ 150

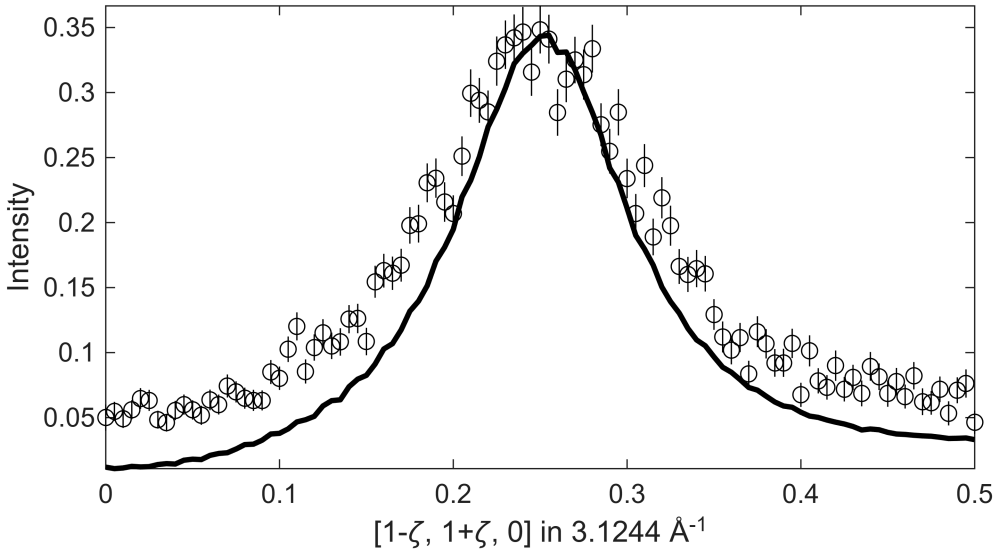
ζ=-0.0025:0.005:0.5025 in [1-ζ, 1+ζ, 0]



Fe_ei401_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1+ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1, -η] , 140 ≤ E ≤ 150

ζ=-0.0025:0.005:0.5025 in [1-ζ, 1+ζ, 0]




```
*****************************
**** En = 155±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 6.6568s    Reduced Chi^2 = 128.1518
 Free parameter values:
         54.12          1.037          35.01         0.1375
```

```
------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 31.3416s    Reduced Chi^2 = 1.5204      Levenberg-Marquardt = 0.1
 Free parameter values:
        45.71         0.9004          34.97        0.03957
------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 56.8697s    Reduced Chi^2 = 1.0153      Levenberg-Marquardt = 0.01
 Free parameter values:
        53.71          1.046          35.42        0.02617
------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 81.8708s    Reduced Chi^2 = 1.0109      Levenberg-Marquardt = 0.001
 Free parameter values:
        54.95          1.072          35.57        0.02493
------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 107.178s    Reduced Chi^2 = 1.0156      Levenberg-Marquardt = 0.0001
 Free parameter values:
        54.97          1.073          35.56        0.02489
 Total time = 112.895s    Reduced Chi^2 = 1.0092      Levenberg-Marquardt = 0.001
 Free parameter values:
        54.97          1.073          35.56        0.02489
------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 137.2343s    Reduced Chi^2 = 1.0056      Levenberg-Marquardt = 0.0001
 Free parameter values:
        54.97          1.073          35.57         0.0249
------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 161.7242s    Reduced Chi^2 = 1.0062      Levenberg-Marquardt = 1e-05
 Free parameter values:
        55.01          1.073          35.57        0.02491
 Total time = 167.9761s    Reduced Chi^2 = 1.0028      Levenberg-Marquardt = 0.0001
 Free parameter values:
        55.01          1.073          35.57        0.02491
------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 193.6986s    Reduced Chi^2 = 1.0096      Levenberg-Marquardt = 1e-05
 Free parameter values:
        54.96          1.073          35.57        0.02492
 Total time = 199.3992s    Reduced Chi^2 = 1.0096      Levenberg-Marquardt = 0.0001
 Free parameter values:
        54.96          1.073          35.57        0.02492
 Total time = 205.8258s    Reduced Chi^2 = 1.0168      Levenberg-Marquardt = 0.001
 Free parameter values:
        54.96          1.073          35.57        0.02492
 Total time = 211.8423s    Reduced Chi^2 = 1.0081      Levenberg-Marquardt = 0.1
 Free parameter values:
        54.98          1.073          35.57         0.0249
 Total time = 217.5497s    Reduced Chi^2 = 1.0091      Levenberg-Marquardt = 10
 Free parameter values:
        55.01          1.073          35.57         0.0249
 Total time = 223.3601s    Reduced Chi^2 = 1.0086      Levenberg-Marquardt = 1000
 Free parameter values:
        55.01          1.073          35.57        0.02491
 *** No improvement in chi-squared over previous iteration ***
 Total time = 223.3674s    Reduced Chi^2 = 1.0028
 Free parameter values:
        55.01          1.073          35.57        0.02491
------------------------------------------------------------------------------
```

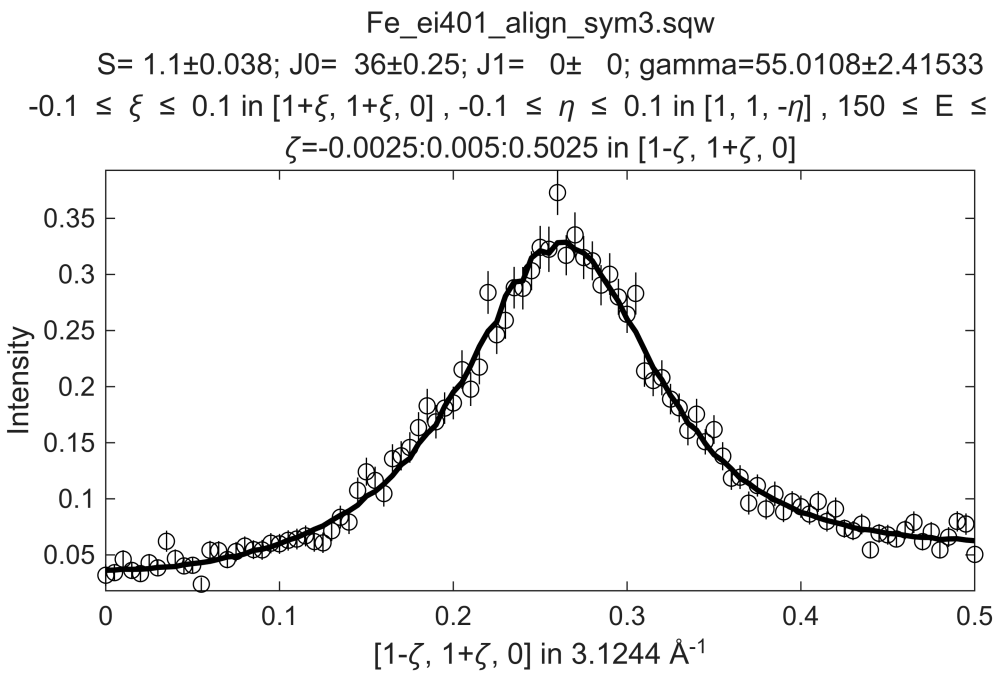
```
Fit converged:

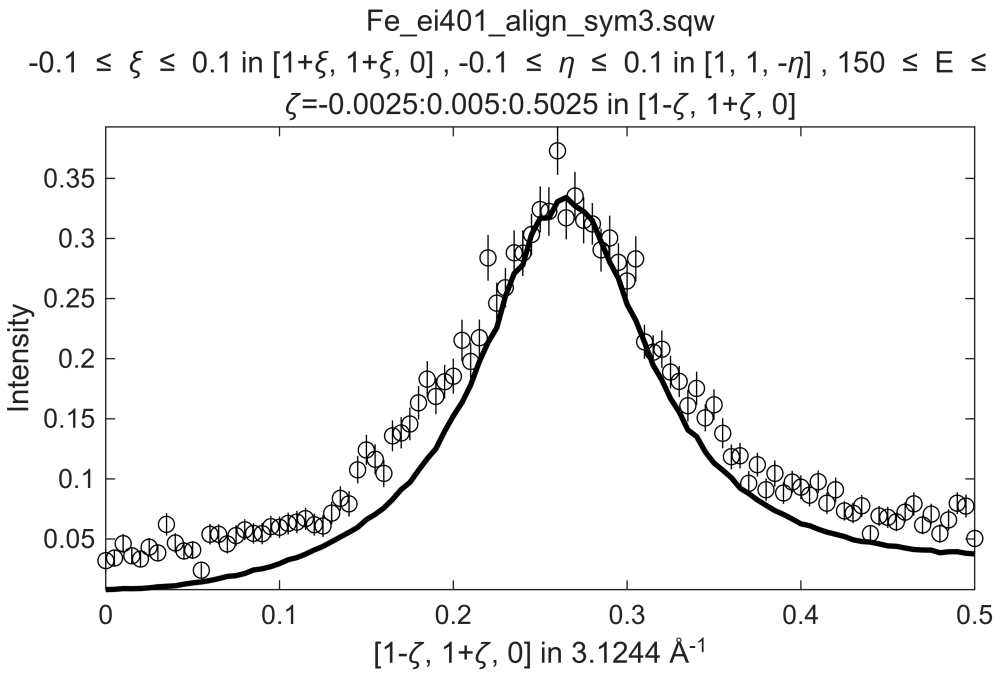
Parameter values (free parameters with error estimates):
-------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          55.01   +/-   2.415
    4          1.073   +/-   0.03778
    5              0
    6          35.57   +/-   0.2523
    7              0
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.02491   +/-   0.002071
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9098    0.7910   -0.9798
    0.9098    1.0000    0.8136   -0.9719
    0.7910    0.8136    1.0000   -0.8467
   -0.9798   -0.9719   -0.8467    1.0000
```






```
*****************************
**** En = 165±5 going to 165
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 4.2786s    Reduced Chi^2 = 84.0686
 Free parameter values:
         55.01          1.073          35.57         0.1282
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 22.1343s    Reduced Chi^2 = 1.449      Levenberg-Marquardt = 0.1
 Free parameter values:
         45.49         0.8605          34.85        0.03682
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 39.3947s    Reduced Chi^2 = 1.1219      Levenberg-Marquardt = 0.01
 Free parameter values:
         53.72           1.02          35.16        0.02201
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 55.2667s    Reduced Chi^2 = 1.1121      Levenberg-Marquardt = 0.001
 Free parameter values:
          54.4          1.043          35.26        0.02084
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 70.3417s    Reduced Chi^2 = 1.1152      Levenberg-Marquardt = 0.0001
 Free parameter values:
          54.4          1.042          35.27        0.02091
 Total time = 75.0058s    Reduced Chi^2 = 1.1154      Levenberg-Marquardt = 0.001
 Free parameter values:
          54.4          1.042          35.27        0.02091
 Total time = 79.5626s    Reduced Chi^2 = 1.1164      Levenberg-Marquardt = 0.01
 Free parameter values:
          54.4          1.042          35.27        0.02091
 Total time = 83.6258s    Reduced Chi^2 = 1.119      Levenberg-Marquardt = 1
 Free parameter values:
         54.41          1.043          35.26        0.02087
 Total time = 87.7853s    Reduced Chi^2 = 1.1222      Levenberg-Marquardt = 100
 Free parameter values:
```

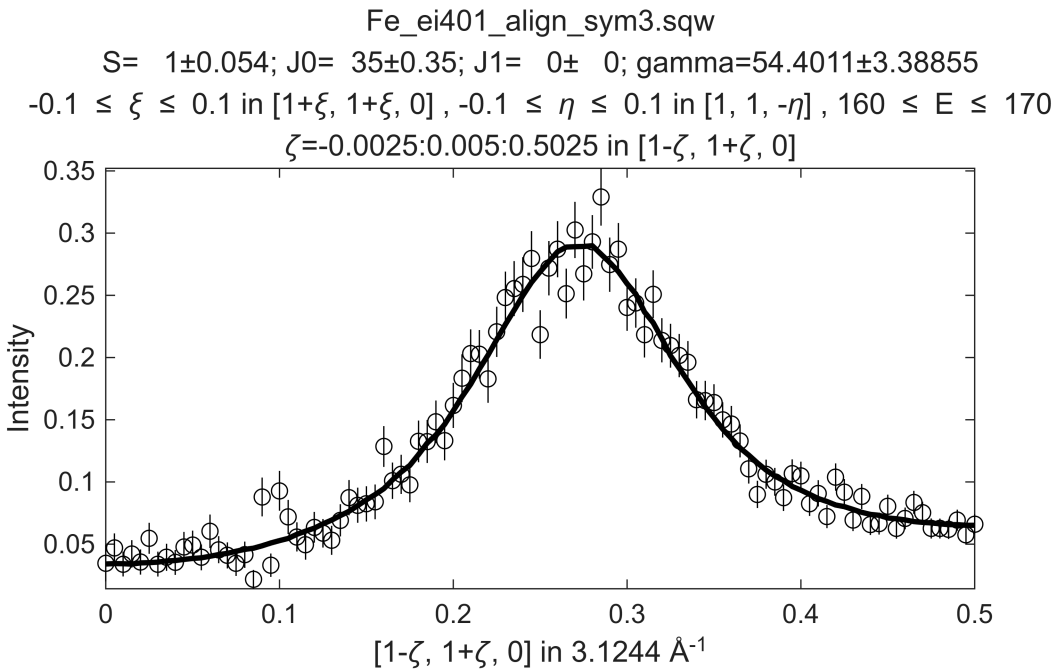
```
          54.4          1.043          35.26        0.02085
 Total time = 92.4083s    Reduced Chi^2 = 1.1116      Levenberg-Marquardt = 10000
 Free parameter values:
          54.4          1.043          35.26        0.02084
------------------------------------------------------------------------------
Fit converged:

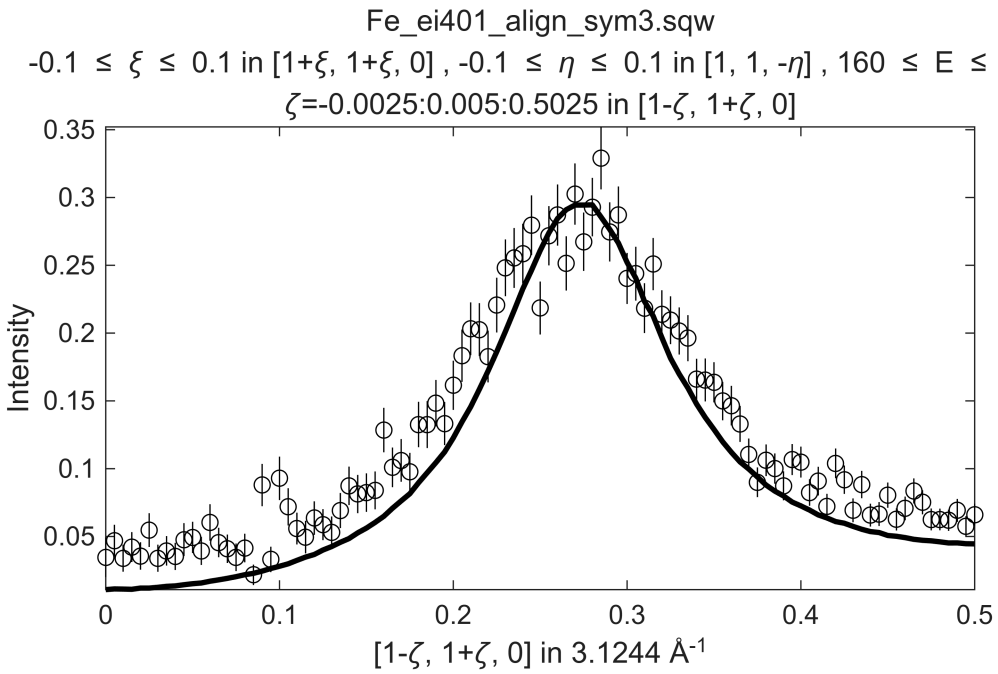
Parameter values (free parameters with error estimates):
---------------------------------------------------------

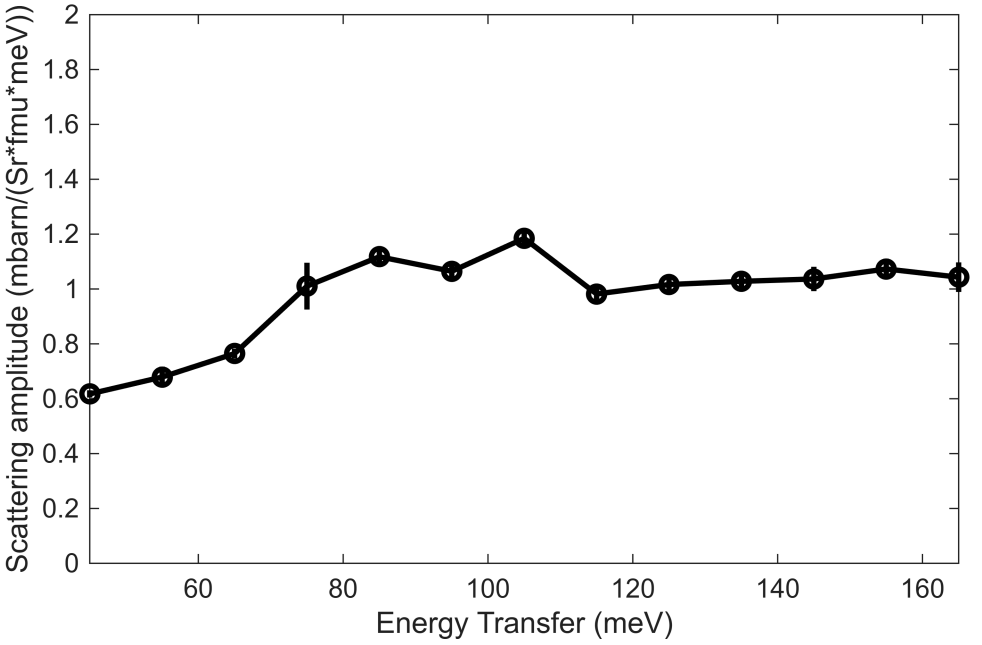
Foreground parameter values:
----------------------------
    1              1
    2              8
    3           54.4   +/-   3.389
    4          1.043   +/-   0.05391
    5              0
    6          35.26   +/-   0.3509
    7              0
    8              0
    9              0
   10              0

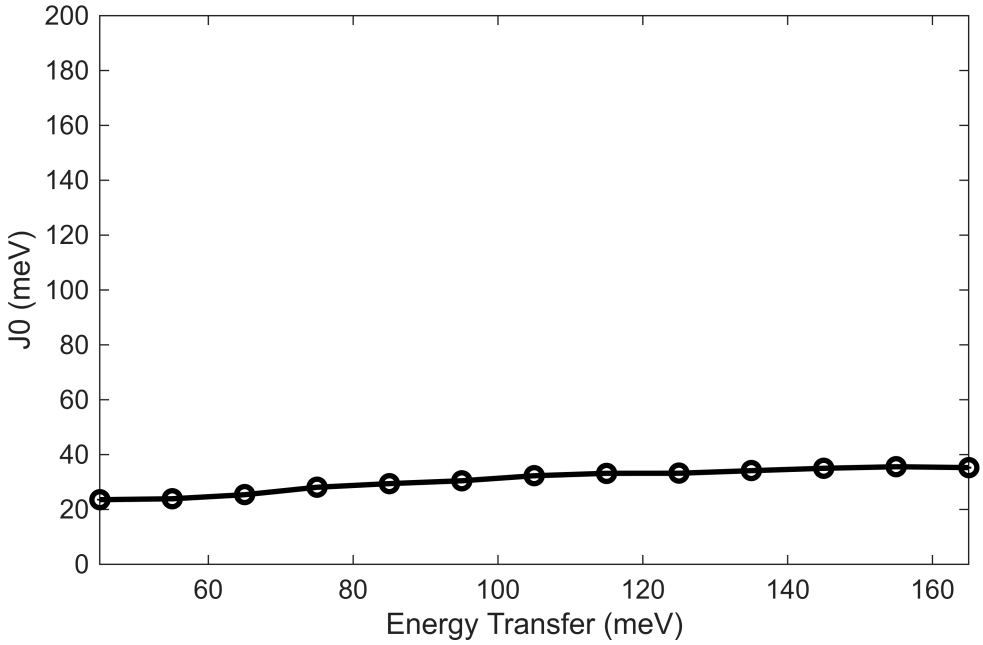

Background parameter values:
----------------------------
    1        0.02084   +/-   0.003372
    2              0
    3              0

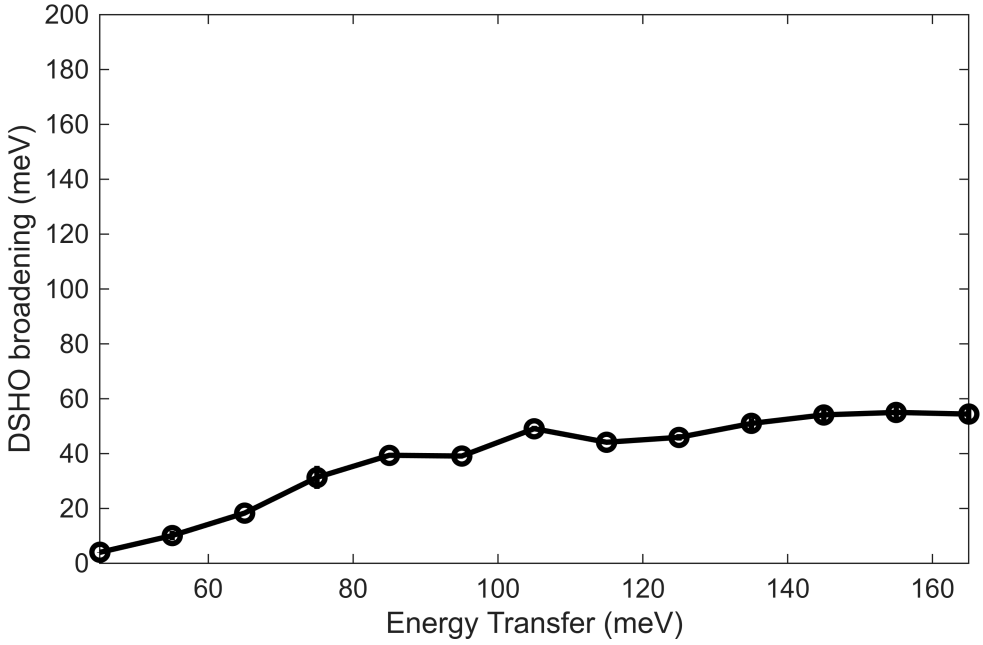

Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9007    0.8190   -0.8038
    0.9007    1.0000    0.7823   -0.9020
    0.8190    0.7823    1.0000   -0.5826
   -0.8038   -0.9020   -0.5826    1.0000
```




Fe_ei401_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1+ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1, -η] , 160 ≤ E ≤ 170
ζ=-0.0025:0.005:0.5025 in [1-ζ, 1+ζ, 0]
Intensity
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0.1
0.2
0.3
0.4
0.5
[1-ζ, 1+ζ, 0] in 3.1244 Å-1


Scattering amplitude (mbarn/(Sr*fmu*meV))
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
60
80
100
120
140
160
Energy Transfer (meV)

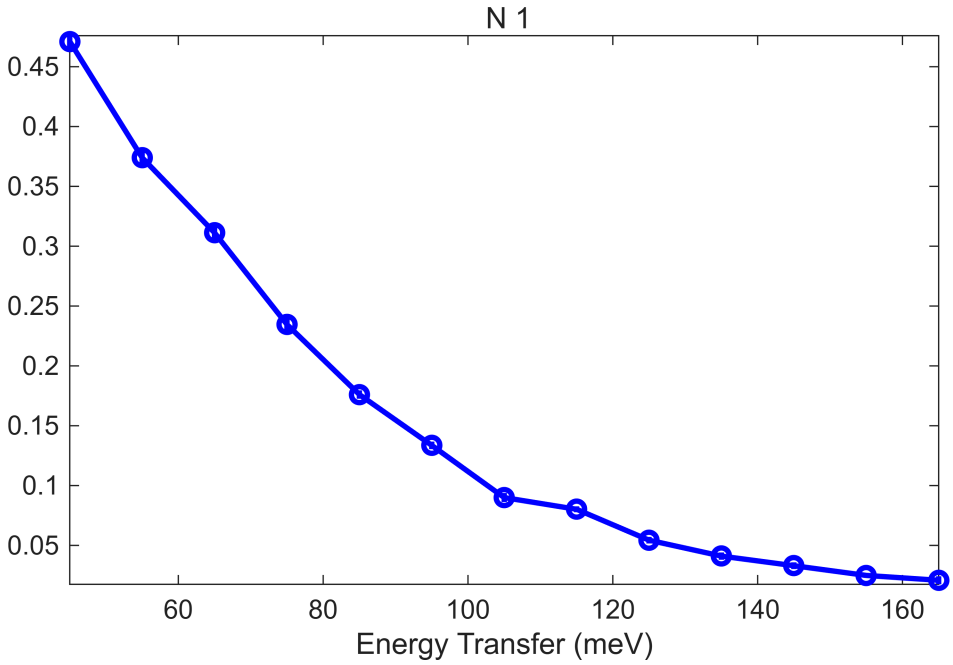




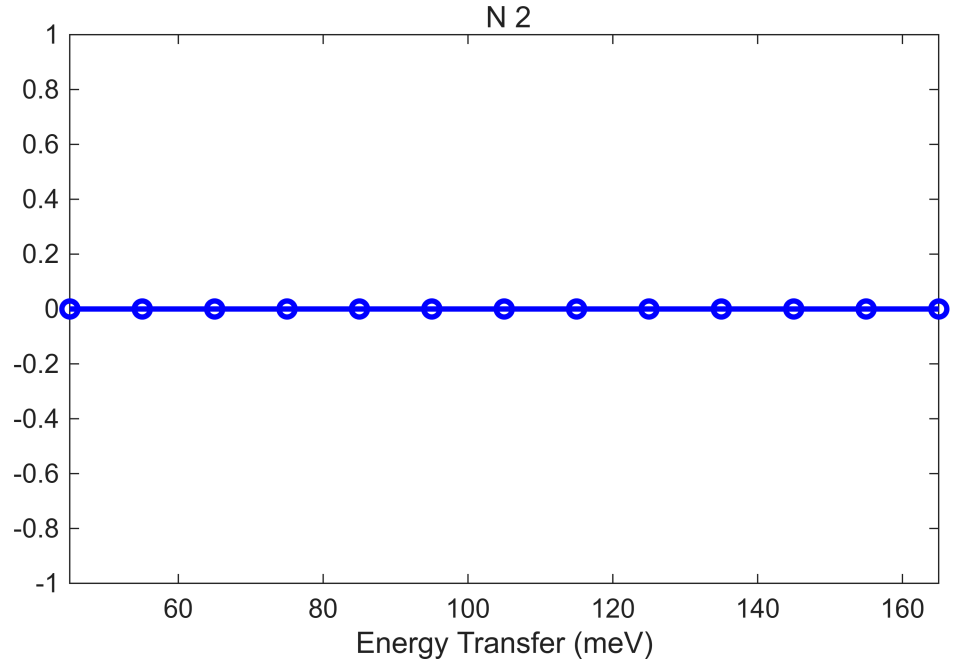

```
cl = NG_FFCutsFits.(selectCuts);
NG_FFCutsFits.(selectCuts) = [cl{:}];

beep
beep
beep
```

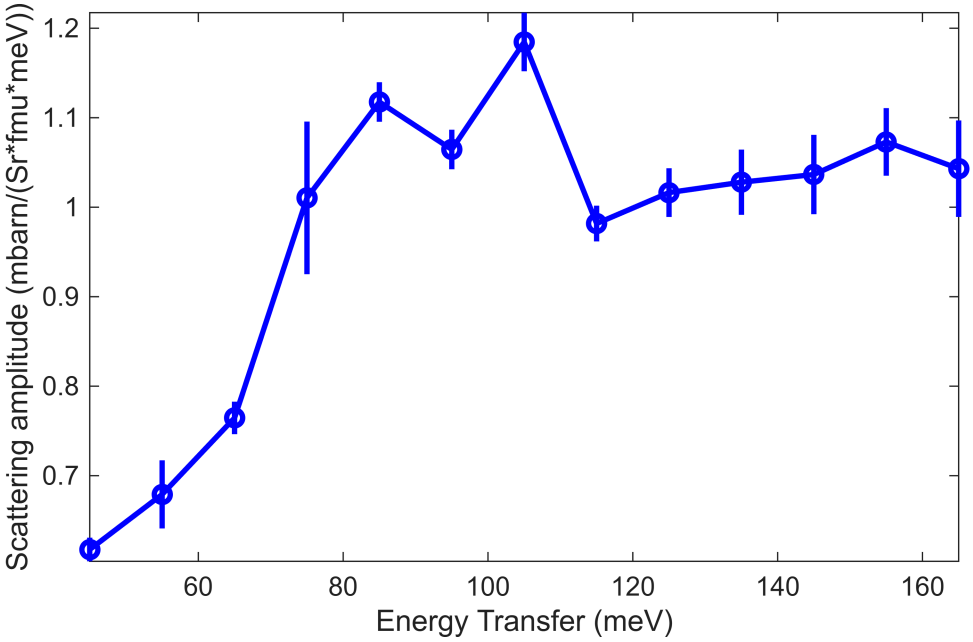
```
all_fit_par = NG_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
acolor b
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en); keep_figure;
```



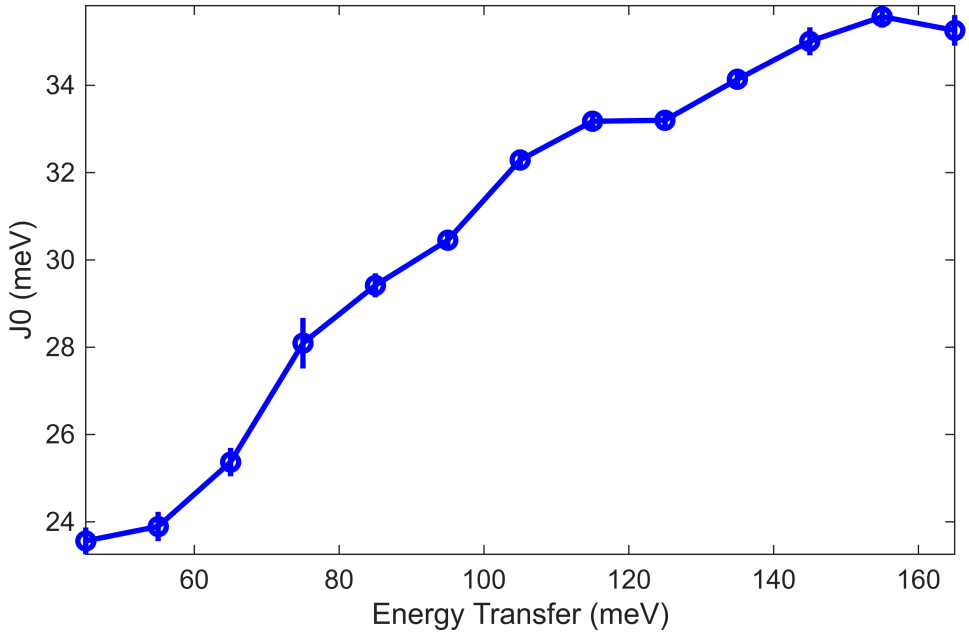

```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



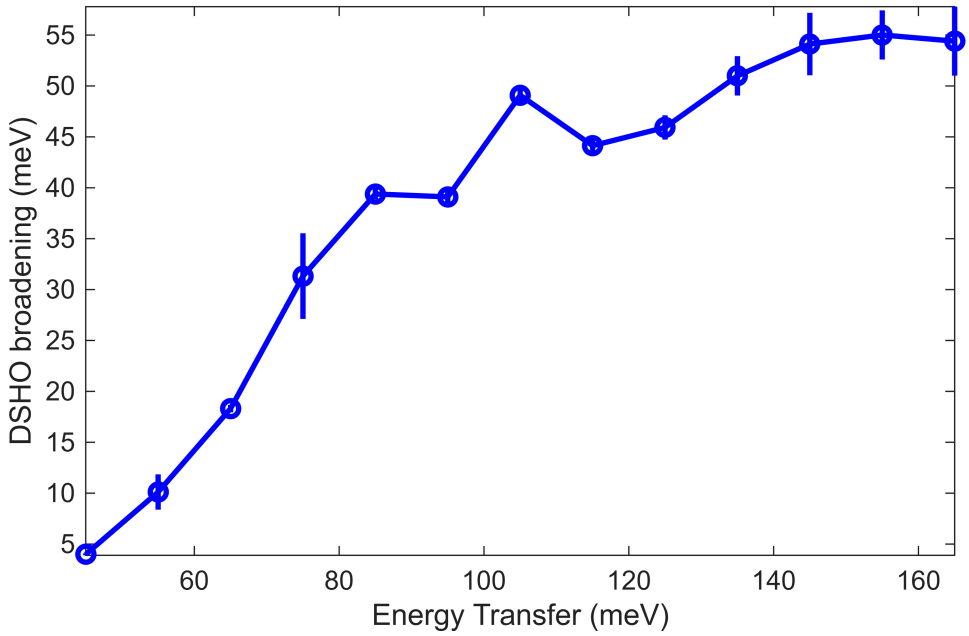

```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure
[S_eff,J0_eff,G_eff] = extract_fit_par(all_fit_par );
pd(S_eff); keep_figure;
```




```
pd(J0_eff);keep_figure
```




```
pd(G_eff); keep_figure;
```




```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_NG.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_NG_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
fit_src_struc = NG_FFCutsFits;
save(CutFitFileNG,"fit_src_struc",'-v7.3')
```